

Co-optimization of multiple virtual power plants considering electricity-heat-carbon trading: A Stackelberg game strategy

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ABSTRACT

With the improvement of the electricity-heat-carbon trading mechanism, it has been a trend for multiple virtual power plants (MVPP) to participate in the market competition. Firstly, in order to mitigate the burden of environmental pollution and balance the conflicting market interests of MVPP, this paper proposes a Stackelberg game strategy for the operator and MVPP considering a ladder-type carbon price mechanism and a dual-compensated demand response (DR) mechanism, which consists of an optimal pricing strategy for the operator and an optimal energy scheduling strategy for each VPP. During the game process, the operator serves as the leader, guiding the energy transactions of MVPP through price optimization, while each VPP acts as the follower, optimizing the internal aggregated units and loads based on the price issued. To solve the Stackelberg game equilibrium, a Kriging metamodel is then adopted to fit the energy scheduling model of the VPP. Finally, several cases are analyzed to verify that the proposed strategy can effectively contribute to the energy complementation and carbon reduction of MVPP.

1. Introduction

With the development of distributed energy resources (DER), the energy structure of the world is gradually shifting from the centralized utilization of large-scale fossil energy to the distributed usage of multi-type energy resources, which will greatly relieve the pressures on energy exploitation and environmental protection [1]. However, due to the small capacity and fluctuating power output of DERs, their large-scale grid connection will pose a series of safety problems. To address the challenge, a virtual power plant (VPP) aggregating DERs, energy storage equipment, and loads is developed, which breaks the limits of geographic location and resource type [2]. As an integrated entity, the VPP can carry out energy dispatch and market trading independently, providing an effective solution to the large-scale grid connection and rational consumption of DERs [3].

Currently, the energy scheduling strategy for a VPP in the context of the market transaction is a hot topic of research. Reference [4] proposed a combined approach for interval and deterministic optimization to solve the VPP scheduling problem. Reference [5] explored a two-stage optimal VPP operational mechanism considering the reserve uncertainty, which includes a novel day-ahead offering strategy and a real-

time dispatching model. Reference [6] established a VPP optimal scheduling model of energy sharing and interaction among various regions, and the model is solved by the compound differential evolution algorithm. Reference [7] proposed a multi-timescale VPP scheduling scheme, including a scenario-based robust optimization for the day-ahead scheduling and two receding horizon optimizations for the close-to-real-time dispatch. With the development of the electricity-heat-carbon trading mechanism, multiple VPPs (MVPP) combining energy production and consumption gradually compete in the market [8]. Driven by the market transaction prices, each VPP, as an independent stakeholder, tends to pursue its maximum interest. Therefore, the current investigated methods for optimizing the scheduling of a single VPP are not applicable to the optimization of MVPP.

In general, the fundamental issue required to be solved for the collaborative optimal scheduling of MVPP is the description of the interaction behavior between the different stakeholders. Game theory is a common method for studying the equilibrium of interests between multiple agents, which is widely used in the economic field [9]. At present, many researchers focus on using game theory to solve the optimal scheduling problems of various market participants. In reference [10], the advertising differential game is introduced to depict the competition among electricity operators in continuous time. In reference

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Nomenclature**Abbreviation**

CEA	Carbon emission allowance
CHP	Combined heat and power supply device
DEL	Dispatchable electric load
DER	Distributed energy resource
DHL	Dispatchable heat load
DL	Dispatchable load
DR	Demand response
ESS	Energy storage system
GB	Gas boiler
GT	Gas turbine
IDR	Incentive-based demand response
IEL	Interruptible electric load
IHL	Interruptible heat load
KKT	Karush-Kuhn-Tucker
LHS	Latin hypercube sampling
MVPP	Multiple virtual power plant
PDR	Price-based demand response
PSO	Particle swarm optimization
PV	Photovoltaics
SEL	Shiftable electric load
SHL	Shiftable heat load
VPP	Virtual power plant
WHRB	Waste heat recovery boiler
WT	Wind turbine

Indices

i	Index of carbon emission source
k	Index of VPP
s_{el}, s_{hl}	Index of electric load and heat load scenario
s_p, s_w	Index of PV and wind power scenario
t	Time interval

Variables

C_{ca}^i	Carbon trading cost of the i th carbon emission source [¥]
$C_{dr,t}^k$	Dual compensation cost of VPPk at time t [¥]
$C_{arc,t}^k$	Carbon subsidy cost of VPPk at time t [¥]
$C_{drp,t}^k$	Energy price compensation cost of VPPk at time t [¥]
$C_{ess,t}^k$	Rental cost of the ESS in VPPk at time t [¥]
$C_{fc,t}^k, C_{fg,t}^k$	Fuel cost of the CHP and GB in VPPk at time t [¥]
$C_{grid,t}$	Electricity purchase cost from the upper grid at time t [¥]
$C_{se,t}^k$	Electricity selling revenue from the operator to VPPk at time t [¥]
$C_{sh,t}^k$	Heat selling revenue from the operator to the VPPk at time t [¥]
$C_{tran,t}^k$	Energy transmission cost between the operator and VPPk [¥]
E_{ce}^k, E_{ge}^k	Carbon emissions of the CHP and GB in VPPk [t]
E_{chp}^k, E_{gb}^k	CEAs of CHP and GB in VPPk [t]
E_{tr}^i	CEAs trading volume of the i th carbon emission source [t]
E_{va}^i	CEAs allocated to the i th carbon emission source [t]
E_{ve}^i	Carbon emissions of the i th carbon emission source [t]
F_{op}	Profit of the operator [¥]
F_{VPPk}	Operation cost of VPPk [¥]
$P_{ae,t}^k, Q_{ah,t}^k$	Actual electric and heat load of VPPk at time t after DR
$P_{ech,t}^k, P_{edis,t}^k$	Power charged and discharged in VPPk at time t [kW]
$P_{ess,t}^k$	Energy stored of ESS in VPPk at time t [kWh]
$P_{gt,t}^k$	Electrical output of the GT in VPPk at time t [kW]
$P_{iel,t}^k, Q_{ihl,t}^k$	Electric and heat load curtailed in VPPk at time t [kW]

$P_{spe,t}^k, Q_{sph,t}^k$	Electric and heat load of VPPk transferred from other time periods to time t [kW]
$P_{sne,t}^k, Q_{snh,t}^k$	Electric and heat load of VPPk transferred from time t to other time periods [kW]
$P_{vppeb,t}^k$	Power purchased by VPPk from the operator [kW]
$P_{vppe,t}^k$	Power sold by VPPk to the operator [kW]
$Q_{gb,t}^k, Q_{whb,t}^k$	Thermal output of the GB and WHRB in VPPk at time t [kW]
$Q_{vpphb,t}^k$	Heat purchased by VPPk from the operator [kW]
$Q_{vpphs,t}^k$	Heat sold by VPPk to the operator [kW]
$\lambda_{vppeb,t}, \lambda_{vppe,t}$	Power purchase and selling price of VPPk at time t [¥/kWh]
$\lambda_{vpphb,t}, \lambda_{vpphs,t}$	Heat purchase and selling price of VPPk at time t [¥/kWh]

Parameters

a_e^k, b_e^k, c_e^k	Cost coefficients of the GT
a_h^k, b_h^k, c_h^k	Cost coefficients of the GB
L_g	Low heating value of the natural gas [kWh/m ³]
m	Total number of MVPP
N_{cd}	Maximum number of the discharge and charge operations
$N_{iel}, N_{ihl}, N_{se}, N_{sh}$	Maximum number of dispatches for IEL, IHL, SEL, and SHL
$P_{el,t}^k$	Electric load of VPPk at time t after stochastic optimization [kW]
P_{el,t,s_d}^k	Forecasted electric load of VPPk in scenario s_{el} at time t [kW]
$P_{ess,min}^k, P_{ess,max}^k$	Minimum and maximum capacity of ESS in VPPk [kWh]
$P_{edis,max}^k, P_{ech,max}^k$	Maximum discharge and charge quantity of ESS in VPPk [kW]
$P_{pv,t}^k$	PV output of VPPk at time t after stochastic optimization [kW]
P_{pv,t,s_p}^k	Forecasted PV output of VPPk in scenario s_p at time t [kW]
$P_{wt,t}^k$	Wind power of VPPk at time t after stochastic optimization [kW]
P_{wt,t,s_w}^k	Forecasted wind power of VPPk in scenario s_w at time t [kW]
$P_{gt,max}^k, P_{pv,max}^k, P_{wt,max}^k$	Maximum power output of the GT, PV, and WT in VPPk [kW]
$P_{vo,max}^k$	Maximum power trading volume of VPPk [kW]
$Q_{hl,t}^k$	Heat load of VPPk at time t after stochastic optimization [kW]
$Q_{hl,t,s_{hl}}^k$	Forecasted heat load of VPPk in scenario s_{hl} at time t [kW]
$Q_{gb,max}^k$	Maximum heat output of the GB in VPPk [kW]
$Q_{vo,max}^k$	Maximum heat trading volume of VPPk [kW]
S	Total number of scenarios
T	Length of the planning horizon [h]
$\alpha_{drie}, \alpha_{drih}, \alpha_{drse}, \alpha_{drsh}$	Compensatory tariff for IEL, IHL, SEL, and SHL [¥/kWh]
α_{rent}^k	Rental cost of ESS in VPPk [¥/kWh]
$\alpha_{te}^k, \alpha_{th}^k$	Conversion coefficients of the VPPk power and heat transmission costs
$\pi_{t,s_d}^k, \pi_{t,s_{hl}}^k, \pi_{t,s_w}^k, \pi_{t,s_p}^k$	Probability of scenario s_{el}, s_{hl}, s_w , and s_p
ν	Benchmark carbon price [¥/t]
μ	Interval length of the ladder-type carbon price [t]
λ_c	Growth rate of the carbon price
$\lambda_{gb,t}$	Power purchase price from the upper grid [¥/kWh]
$\lambda_{gs,t}$	Power selling price to the upper grid [¥/kWh]

$\lambda_{hs,ave}, \lambda_{hb,ave}$	Average heat selling and purchase price [¥/kWh]	η_{dis}, η_{ch}	Discharging and charging efficiency of ESS
ε_c	Carbon emission coefficient for unit coal-fired generation [kg/kWh]	$\tau_{se}^k, \tau_{sh}^k, \tau_{iel}^k, \tau_{ihl}^k$	Maximum proportion of SEL, SHL, IEL, and IHL in VPPk
ε_g	Carbon emission coefficient for unit natural gas [kg/m ³]	$\varphi_{gtt,max}^k, \varphi_{gtd,max}^k$	Maximum ramping and descending rate of the GT in VPPk [kW/h]
σ_c	Share of the coal-fired generation in the superior power purchase	$\varphi_{gbu,max}^k, \varphi_{gbd,max}^k$	Maximum ramping and descending rate of the GB in VPPk [kW/h]
χ_e, χ_h	CEA coefficient for unit electricity and heat [kg/kWh]		
$\eta_{gt}, \eta_{gb}, \eta_{whb}$	Working efficiency of the GT, GB, and WHRB		

[11], a market mechanism for real-time reactive power ancillary service market based on the Stackelberg game model is proposed considering voltage-apparent power coupled subsystems. In reference [12], the interaction between consumers and the operator is modeled as a Stackelberg game to analyze the impact of real-time electricity. By analyzing the game objectives and strategies of the multi-agent system, the unique Stackelberg equilibrium solution of the hierarchical Stackelberg game is proved theoretically in reference [13]. To study the regulatory performance of the dynamic electricity price mechanism adopted by the power operator, a non-cooperative Stackelberg model is constructed based on game theory in reference [14]. These studies have confirmed that the theory plays an important role in power trading to address the interaction of interests between MVPP. However, current studies mainly focus on operator-led electricity stakeholders and rarely consider entities with multi-energy coupling properties. Hence, the MVPP game considering electrothermal coupling and multi-energy trading is required to be studied in depth.

As the game theory involves a multi-layer model optimization, the solution of the game model of MVPP can be converted into a large-scale optimization problem for complex systems [15]. Generally speaking, the heuristic intelligence algorithm and the analytical approach based on Karush-Kuhn-Tucker (KKT) conditions are the two main methods for solving the game equilibrium solution. The heuristic intelligent algorithm involves frequent calls to models, which will reduce the solving speed [16]. Whereas the analytical approach based on KKT conditions requires the invocation of all parameter information, which will pose privacy issues [17]. In addition, KKT conditions are suitable for seeking the optimal solution of a convex optimization problem, which imposes a convex requirement on the objective function and constraints of the model [18]. To address these problems, it is essential to simplify the complex game model. Due to the good fitting and error estimation capability for non-linear models, the Kriging metamodel is used as an approximate alternative to the complicated model [19]. Referring to the successful application of the Kriging metamodel in the optimal power flow problem solving [20] and the economic dispatch problem solving considering the wind uncertainty [21], it would be introduced to reduce the difficulty of the game equilibrium solution solving. Moreover, as the fit accuracy of the metamodel directly affects the game results, a correction mechanism of the Kriging metamodel should be further researched to improve the fitting ability.

With the development of the carbon market, carbon trading is considered to be an essential tool for balancing the environment and economy [22]. Researches on the carbon trading of the VPP are ongoing. Based on the optimization of cooling, heating, and electricity load and the ladder-type carbon trading mechanism, reference [23] established an optimal scheduling model with a carbon capture device. Reference [24] introduced a carbon trading mechanism and a carbon emission calculation model and proposed an optimal scheduling strategy for a VPP considering the carbon trading cost. Reference [6] incorporated the cost of carbon trading into the economic operation model of a VPP with carbon capture equipment and analyzed the impact of carbon trading prices on the VPP operation. Reference [25] explored a ladder-type carbon trading mechanism and studied the impact of a ladder-type carbon price on system operation profits and carbon emissions. Based on the life cycle assessment for the energy chains, the normalized carbon

emission coefficient and the ladder-type carbon trading mechanism considering the reward and punishment are adopted in reference [26] to calculate the carbon trading cost. Reference [27] examined factors affecting carbon pricing, including environmental, social, political, and economic variables. Existing research on carbon trading concentrate on the innovation of transaction mechanisms, the tracking of carbon footprints, and the analysis of pricing influences, exploring the impact of carbon trading on systemic carbon emissions and operating costs. However, the current studies mainly investigate a single VPP and rarely examine the joint carbon trading of MVPP, especially the impact of energy interactions between MVPP on carbon trading.

To manage the load distribution within the VPP, a demand response (DR) mechanism is commonly adopted to compensate the energy consumers for reducing or transferring the loads during peak hours of energy usage [28]. Existing researches focus on analyzing the impact of DR on costs and load distributions. Reference [29] suggested that implementing DR would decrease the total operating costs of VPP. Reference [30] indicated that the DR scheme induced more load reductions, which mitigated critical load and decreased VPP operation costs. Reference [31] indicated that the price-based DR could realize peak load shifting, and the incentive-based DR could reduce the operation cost. In reference [32], the energy and reserve scheduling with and without DR participants is addressed in detail via comparison. As a component of the VPP dispatch strategy, a sequential DR scheme was proposed in reference [33] to reduce both ramping rate and peak power at the aggregated level. Reference [34] reduced the penalty costs in day-ahead and real-time scheduling through the balance of spinning reserves with DR services. However, few studies examine the impact of DR on the carbon trading of the VPP from the load side. Moreover, current researches on DR are relatively simple, with a single compensation mechanism, and the impact of DR on the carbon emissions of MVPP is rarely analyzed.

In response to the above issues, this paper proposes a Stackelberg game strategy for the operator and MVPP considering a ladder-type carbon price mechanism and a dual-compensated DR mechanism, which consists of an optimal pricing strategy for the operator and an optimal energy scheduling strategy for the VPP. To solve the Stackelberg game equilibrium solution, a corrected Kriging metamodel is adopted. The main contributions of this paper are summarized as follows.

- (1) Considering the electrothermal coupling for MVPP, a Stackelberg game model is developed to coordinate the energy transactions between the operator and MVPP, which will be beneficial for energy consumption and the complementation of VPP.
- (2) To improve the ability of the Kriging metamodel to fit the MVPP model, this paper proposes a dynamic update algorithm of sampling points to correct the Kriging metamodel.
- (3) Based on a ladder-type carbon price mechanism, the impact of MVPP energy interactions driven by the Stackelberg game on carbon trading is researched, and the carbon reduction potential is further explored.
- (4) Unlike the traditional DR mechanism with a single price compensation, this paper establishes a dual-compensated DR mechanism considering the energy price compensation and the carbon subsidy and investigates the contribution of DR to carbon emission reduction.

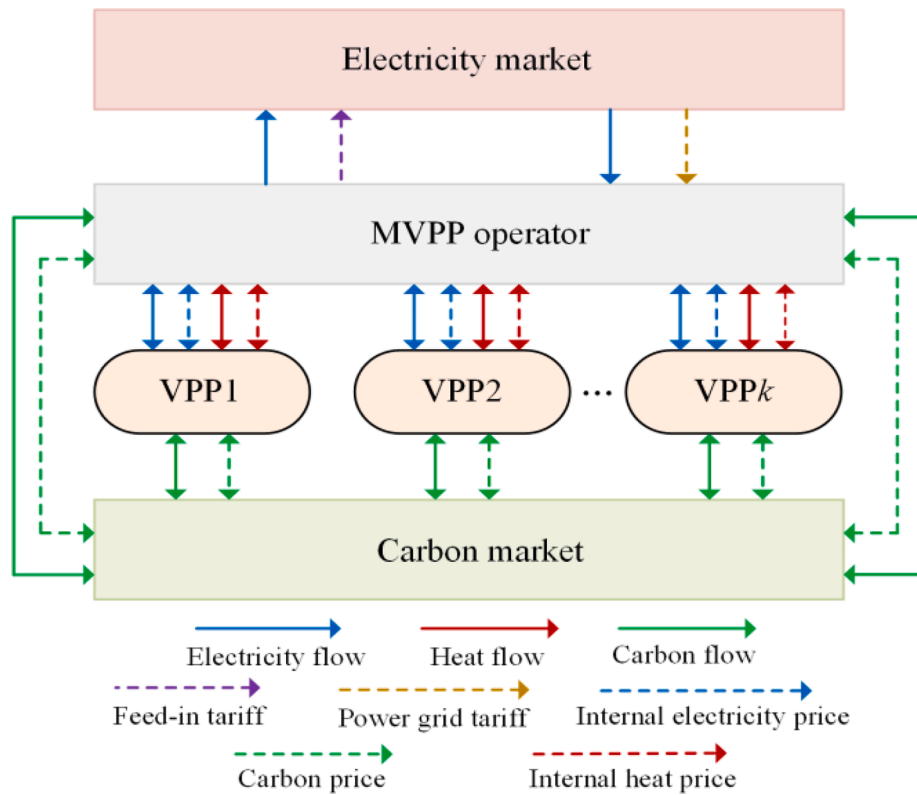


Fig. 1. The structure of the operator and MVPP.

The remainder of the paper is organized as follows: [Section 2](#) introduces the operational characteristics of the operator and MVPP, including the Stackelberg game structure, the framework of the ladder-type carbon price mechanism, and the framework of the dual-compensated DR mechanism. [Section 3](#) establishes mathematic models of the operator and MVPP including objective functions and constraints, and proves the Nash equilibrium of the Stackelberg game model. [Section 4](#) introduces the solving method for the established model. The case study and result analysis are presented in [section 5](#). [Section 6](#) draws the conclusion.

2. Operational characteristics for the operator and MVPP

In this paper, the VPP consists of wind turbines (WT), photovoltaics (PV), gas turbines (GT), waste heat recovery boilers (WHRB), heat exchangers (HE), gas boilers (GB), energy storage systems (ESS), and dispatchable loads (DL), so as to coordinate the cleanliness of renewable energy generation and the controllability of fossil energy supply. Among them, on the basis of the electrical and thermal consumption demands from the energy users, the DL can be further divided into the dispatchable electric loads (DEL) and the dispatchable heating loads (DHL). In addition, the GT, WHRB, and HE comprise a combined heat and power (CHP) supply device. While the GT is generating electricity, the heat in the discharged cooling water and flue gas can be recycled through the WHRB, and together with the heat generated by the GB, the combined heat can be supplied to users through the HE.

Due to the profit-driven nature of market conditions and different characteristics of resource endowments, the internal energy supply, and energy consumption are often unbalanced for each VPP, which presents energy shortages or energy surpluses during some periods. Consequently, an operator is introduced in this paper to coordinate the complementarity of the unbalanced energy resources among MVPP. Drawing on the mechanism of the European electricity market operation [35], the structure of MVPP and the operator is illustrated in [Fig. 1](#). As

shown in [Fig. 1](#), each VPP, as an independent entity, reasonably arranges its internal resource scheduling under the electricity and heat trading price issued by the operator, as well as the forecast information of load and renewable energy, and carries out the transaction of the shortfall or surplus energy with the operator. Acting as an intermediary, the operator on the distribution side aggregates energy trading information from MVPP and then makes the difference by centralizing the purchase and sale of electricity in the electricity market at the power grid tariff and feed-in tariff. In addition, both the operator and MVPP participate in the carbon market transaction simultaneously.

2.1. The Stackelberg game structure

For the energy trading process between the operator and MVPP, the operator prioritizes issuing the pricing strategy based on the energy demand of each VPP, and the VPP then formulates the energy scheduling strategy under the guidance of the price signal. On the contrary, the energy dispatch decisions of MVPP will affect the energy transactions with the operator, which can in turn lead to the adjustment in the pricing decision of the operator. The energy trading process between the operator and MVPP is sequential and interactive, which is in line with the dynamic game of master-slave recursive structure, so the Stackelberg game is adopted in this paper to analyze the interaction between the stakeholders.

In general, the participants, strategies, and objective functions are the three basic elements of a game model, and the framework of the Stackelberg game is illustrated in [Fig. 2](#). As shown in [Fig. 2](#), the operator and MVPP act as participants, taking on the roles of the leader and followers respectively during the process of Stackelberg game. Driven by the objective functions of maximizing the operator's profit and minimizing the VPP operating cost, the corresponding game strategies are formulated, which can be divided into the pricing strategy for the operator and the energy scheduling strategy for each VPP. The pricing strategy involves the purchase and sale prices for the electricity and heat

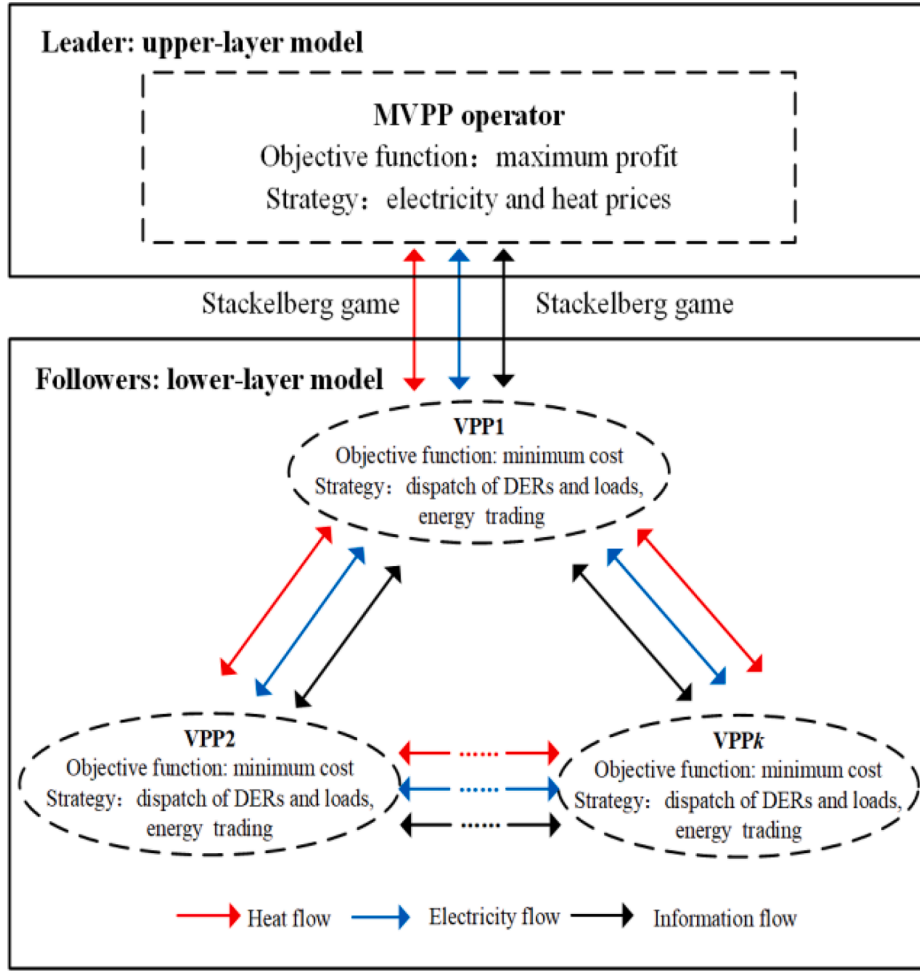


Fig. 2. The Stackelberg game structure.

traded with MVPP, whereas the energy scheduling strategy addresses the dispatch of DERs and loads, as well as energy transactions with the operator.

2.2. Framework of the ladder-type carbon price mechanism

Considering the operating guidelines of the Chinese carbon market, the free carbon emission allowances (CEA) are allocated to each carbon emission source through a baseline method, the allocation basis of which is distinguished by the electrical and thermal energy. In general, when the actual carbon emission is lower than the allocated allowance, the remaining allowance can be sold in the carbon trading market, and conversely, it is required to be purchased. In this paper, the energy production from the CHP and GB within MVPP, and the upper grid power purchase by the operator constitute the main sources of carbon emissions. The process of carbon trading involving a coupled electricity-heat-carbon relationship is illustrated in Fig. 3.

Contrary to traditional carbon trading with a fixed carbon price, the ladder-type pricing mechanism considering the reward and punishment is developed in this paper for the enthusiasm of the carbon reduction and the constraint on carbon emission. According to the mechanism, several CEA purchase and sale bands are divided, and as a penalty, the carbon price for the corresponding interval rises by a certain percentage with the increase of the CEA volume to be purchased. Similarly, the excess CEA can be sold under different sale bands with proportionally increasing reward prices. Therefore, for the operator and MVPP, the carbon trading cost based on the ladder-type carbon price mechanism is formulated as follows.

$$E_{tr}^i = E_{ve}^i - E_{va}^i \quad (1)$$

$$C_{ca}^i = \begin{cases} v(1 + \lambda_c)\mu - v(1 + 2\lambda_c)(E_{tr}^i + \mu) & E_{tr}^i < -\mu \\ v(1 + \lambda_c)E_{tr}^i & -\mu \leq E_{tr}^i < 0 \\ vE_{tr}^i & 0 < E_{tr}^i \leq \mu \\ v\mu + v(1 + \lambda_c)(E_{tr}^i - \mu) & \mu < E_{tr}^i \leq 2\mu \\ v(2 + \lambda_c)\mu + v(1 + 2\lambda_c)(E_{tr}^i - 2\mu) & 2\mu < E_{tr}^i \leq 3\mu \\ v(3 + 3\lambda_c)\mu + v(1 + 3\lambda_c)(E_{tr}^i - 3\mu) & E_{tr}^i > 3\mu \end{cases} \quad (2)$$

$$i \in \{\text{operator, VPP1, VPP2, ..., VPPk}\}$$

The relationship between the ladder-type carbon price and carbon trading volume can be visualized in Fig. 4. For the operator and MVPP, the positive interval indicates that additional CEA should be purchased to meet the carbon emission demand, whereas the negative one reflects that the surplus CEA can be sold for a profit.

2.3. Framework of the dual-compensated DR mechanism

In order to guide energy consumers within the VPP to transfer or reduce part of their loads during peak hours of electricity and heat usage, the DR program for electric and thermal loads is applied in this paper, which can be divided into incentive-based demand response (IDR) and price-based demand response (PDR). According to the two dispatch forms of DR, the DEL and DHL can be further divided into the

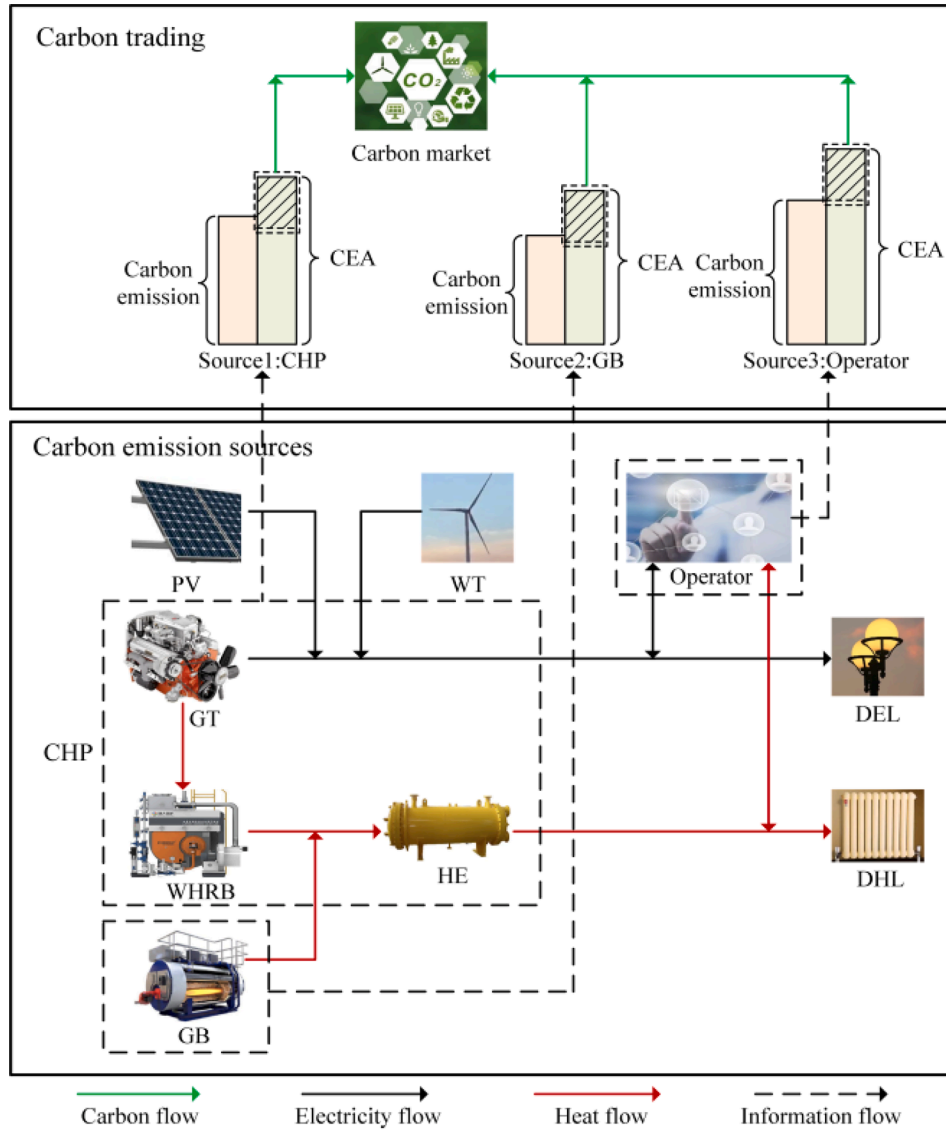


Fig. 3. The process of carbon trading.

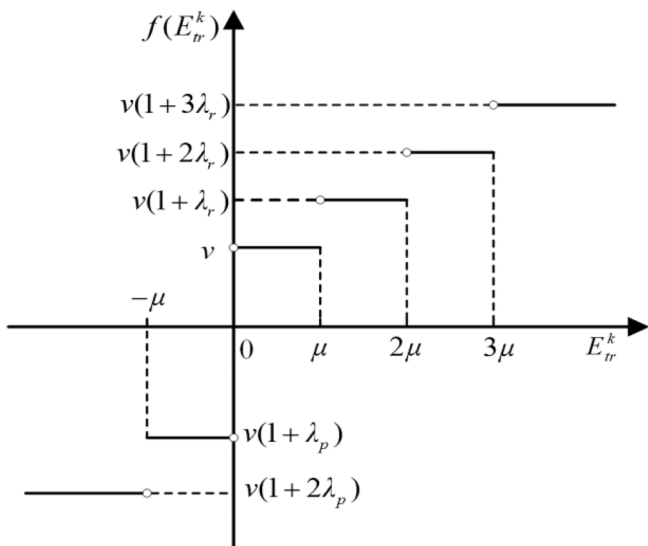


Fig. 4. The ladder-type carbon price mechanism.

shiftable electric load (SEL), shiftable heat load (SHL), interruptible electric load (IEL), and interruptible heat load (IHL). To be more specific, the SEL and SHL allow for spontaneous load shifting in response to price fluctuations, whereas the IEL and IHL reduce their loads with the incentive issued by the VPP.

Due to the interactivity of the energy supply and consumption in the VPP, the adjustment of energy use strategies under the DR will also influence the output scheme of the energy supply equipment, thus affecting the carbon trading of the VPP. Therefore, considering the coupling of electricity, heat, and carbon emissions, this paper proposes a dual compensation mechanism for the DEL and DHL that combines the energy price compensation and the carbon subsidy, so as to investigate the impact of load regulation capability on the energy dispatch and carbon emissions. Generally speaking, in order to guide users to actively reduce energy consumption and adjust energy use periods during peak load hours, energy compensation prices for SEL, SHL, IEL, and IHL are set respectively by the VPP. Then, based on CEA coefficients for unit electricity and heat, the curtailment volume of electric and thermal loads in response to compensation prices are converted into carbon emission reductions, which are dually compensated in the form of carbon subsidies according to the benchmark carbon price.

3. Stackelberg game model for the operator and MVPP

In this paper, the Stackelberg game model includes the upper-layer optimal pricing model for the operator and the lower-layer optimal energy scheduling model for the VPP. The optimal dispatch problem of the operator and MVPP can be described as the optimization of two-layer objective functions while satisfying all constraints.

3.1. Optimal pricing model for the operator

3.1.1. Objective function

As a leader, the MVPP operator coordinating the integrated electricity-heat-carbon trading sets the transaction price with each VPP for its maximum profit, which is formulated as (3)–(7). The revenue of the operator comes from the energy sales to MVPP, and the cost originates in the energy and CEA purchase from the upper grid and the carbon market respectively, as well as the energy transmission with each VPP. In addition, it is worth noting that the cost of energy transmission is shared equally between the operator and each VPP, the derivation of which is detailed in Appendix A.

$$\max F_{op} = \sum_{t=1}^T \sum_{k=1}^m (C_{se,t}^k + C_{sh,t}^k - C_{tran,t}^k - C_{grid,t}^k) - C_{ca}^{operator} \quad (3)$$

$$C_{se,t}^k = \lambda_{vppcb,t} P_{vppcb,t}^k - \lambda_{vppes,t} P_{vppes,t}^k \quad (4)$$

$$C_{sh,t}^k = \lambda_{vpphb,t} Q_{vpphb,t}^k - \lambda_{vpphs,t} Q_{vpphs,t}^k \quad (5)$$

$$C_{tran,t}^k = \frac{1}{2} (\alpha_{te}^k |P_{vppcb,t}^k - P_{vppes,t}^k|^2 + \alpha_{th}^k |Q_{vpphb,t}^k - Q_{vpphs,t}^k|) \quad (6)$$

$$C_{grid,t}^k = \lambda_{gb,t} \max[\sum_{k=1}^m (P_{vppcb,t}^k - P_{vppes,t}^k), 0] + \lambda_{gs,t} \min[\sum_{k=1}^m (P_{vppcb,t}^k - P_{vppes,t}^k), 0] \quad (7)$$

In addition, it is assumed that the power purchased by the operator is partially generated by the coal-fired unit. Based on the carbon trading mechanism, the actual carbon emission and the allocated CEA of the operator are formulated as (8) and (9), and the carbon trading cost can be calculated by the formula (1) and (2).

$$E_{ve}^{operator} = \varepsilon_c \sigma_c \sum_{t=1}^T \max[\sum_{k=1}^m (P_{vppcb,t}^k - P_{vppes,t}^k), 0] \quad (8)$$

$$E_{va}^{operator} = \chi_e \sum_{t=1}^T \max[\sum_{k=1}^m (P_{vppcb,t}^k - P_{vppes,t}^k), 0] \quad (9)$$

3.1.2. Constraints

In order to prevent MVPP from trading with the upper grid directly without coordination from the operator, it is necessary to ensure that the internal energy selling price set by the operator is lower than the market price, and similarly, the internal energy purchase price is higher than the market price. The constraints are as follows.

$$\begin{cases} \lambda_{gs,t} \leq \lambda_{vppes,t} \leq \lambda_{vppcb,t} \leq \lambda_{gb,t} \\ \lambda_{hs,ave} \leq \lambda_{vpphs,t} \leq \lambda_{vpphb,t} \leq \lambda_{hb,ave} \end{cases} \quad (10)$$

3.2. Optimal energy scheduling model for the VPP

3.2.1. Objective function

Based on the internal energy prices issued by the operator, the output of units including CHP, WT, PV, and ESS, and the distribution of aggregated loads are optimized by the VPP. Therefore, the minimum operating cost of the VPP derived from the energy purchase and sale, internal unit operation, DR compensation, energy transmission, and carbon trading is taken as the objective function. It can be formulated as

follows.

$$\min F_{VPPk} = \sum_{t=1}^T (C_{se,t}^k + C_{sh,t}^k + C_{fc,t}^k + C_{fg,t}^k + C_{ess,t}^k + C_{dr,t}^k + C_{tran,t}^k) - C_{ca}^k \quad (11)$$

As the main energy supply equipment, the CHP realizes the conversion and distribution of electricity and heat through the GT, HE, and WHRB. Among them, the HE and WHRB are auxiliary equipment, hence fuel costs of them are not considered. On the basis of the working condition characteristics of the GT and GB, the relationship between their electrical and thermal power outputs and the fuel cost is formulated in the form of a quadratic function as follows.

$$C_{fc,t}^k = a_e^k (P_{gt,t}^k)^2 + b_e^k P_{gt,t}^k + c_e^k \quad (12)$$

$$C_{fg,t}^k = a_h^k (Q_{gb,t}^k)^2 + b_h^k Q_{gb,t}^k + c_h^k \quad (13)$$

The ESS in this paper is leasable, and its rental cost is associated with the stored energy, which is formulated as follows.

$$C_{ess,t}^k = P_{ess,t}^k \alpha_{rent}^k \quad (14)$$

According to the dual compensation DR mechanism for electrical and thermal loads proposed in this paper, the economic compensations for the responding electrical and thermal loads are as follows, including the energy price compensation cost and the carbon subsidy cost.

$$\begin{cases} C_{dr,t}^k = C_{drp,t}^k + C_{drc,t}^k \\ C_{drp,t}^k = P_{iel,t}^k \alpha_{drie} + Q_{ihl,t}^k \alpha_{drih} + |P_{sne,t}^k - P_{spe,t}^k| \alpha_{drse} + |Q_{snh,t}^k - Q_{sph,t}^k| \alpha_{drsh} \\ C_{drc,t}^k = \nu \chi_e P_{iel,t}^k + \nu \chi_h Q_{ihl,t}^k \end{cases} \quad (15)$$

According to the baseline method mentioned above, the CEAs for the electricity and heat supply are calculated as follows.

$$\begin{cases} E_{va}^{VPPk} = E_{chp}^k + E_{gb}^k \\ E_{chp}^k = \chi_e \sum_{t=1}^T P_{gt,t}^k + \chi_h \sum_{t=1}^T Q_{whb,t}^k \\ Q_{whb,t}^k = \frac{P_{gt,t}^k (1 - \eta_{gt})}{\eta_{gt}} \eta_{whb} \\ E_{gb}^k = \chi_h \sum_{t=1}^T Q_{gb,t}^k \end{cases} \quad (16)$$

Correspondingly, through the carbon tracking of the outputs, the actual carbon emissions from the gas-fired units within the VPP are formulated as (17). Then, combined with the CEAs obtained in (16), the carbon trading cost of the VPP can be calculated by the formula (1) and (2).

$$\begin{cases} E_{ve}^{VPPk} = E_{ce}^k + E_{ge}^k \\ E_{ce}^k = \varepsilon_g \sum_{t=1}^T \left(\frac{P_{gt,t}^k}{\eta_{gt} L_g} + \frac{Q_{whb,t}^k}{\eta_{whb} L_g} \right) \\ E_{ge}^k = \varepsilon_g \sum_{t=1}^T \frac{Q_{gb,t}^k}{\eta_{gb} L_g} \end{cases} \quad (17)$$

3.2.2. Constraints

- (1) Operational constraints of the combined heat and power equipment.

As a major component of the CHP equipment, the GT operation is jointly restricted by the output and ramping-descending rate, which is formulated as follows.

$$\begin{cases} 0 \leq P_{gt,t}^k \leq P_{gt,\max}^k \\ -\varphi_{gt,\max}^k \leq P_{gt,t+1}^k - P_{gt,t}^k \leq \varphi_{gtu,\max}^k \end{cases} \quad (18)$$

(2) Operational constraints of the gas boiler.

Similar to the constraint of the CHP, constraint (19) restricts the hourly output, and the ramping and descending rate.

$$\begin{cases} 0 \leq Q_{gb,t}^k \leq Q_{gb,\max}^k \\ -\varphi_{gb,\max}^k \leq Q_{gb,t+1}^k - Q_{gb,t}^k \leq \varphi_{gbu,\max}^k \end{cases} \quad (19)$$

(3) Charge/discharge constraints of the energy storage system.

For the ESS, the energy stored at time t is determined by that at the time $t-1$, and the energy storage should be balanced at the beginning and end of a scheduling period. Its charging and discharging status, which cannot be performed simultaneously, is controlled to stabilize the uncertainty of renewable energy outputs. To protect the service life of the battery, the maximum quantity and frequency for the charge and discharge are restricted. The constraints are formulated as follows.

$$\begin{cases} P_{ess,t}^k = P_{ess,t-1}^k - \frac{P_{edis,t-1}^k}{\eta_{dis}} + P_{ech,t-1}^k \eta_{ch} \\ P_{ess,t=0}^k = P_{ess,t=T}^k \\ P_{ess,\min}^k \leq P_{ess,t}^k \leq P_{ess,\max}^k \\ 0 \leq P_{edis,t}^k \leq U_{dis,t}^k P_{edis,\max}^k \\ 0 \leq P_{ech,t}^k \leq U_{ch,t}^k P_{ech,\max}^k \\ U_{dis,t}^k + U_{ch,t}^k \leq 1 \\ \sum_{t=1}^T (U_{dis,t}^k + U_{ch,t}^k) \leq N_{ed} \end{cases} \quad (20)$$

where $U_{dis,t}^k$ and $U_{ch,t}^k$ are binary variables determining the state of charge and discharge of ESS, 1 reflects the ESS is charged or discharged, and 0 means not working.

(4) Output constraints of the renewable energy.

As the forecasted value of loads, wind power, and photovoltaic within the VPP are uncertain, a stochastic optimization method is adopted to characterize the randomness of the energy usage and renewable energy power generation, including the Latin hypercube sampling (LHS)-based scenario generation and the forward selection technology-based scenario elimination [36]. Specifically, for each uncertainty, 1000 scenarios are generated and 10 typical scenarios are selected in this paper, and the probability of each corresponding scenario can be obtained. With stochastic optimization, the probability of each scenario is multiplied by the corresponding forecasted value and summed to calculate the final value of the uncertainty per hour. Constraint (21) restricts the output of renewable energy.

$$\begin{cases} 0 \leq P_{wt,t}^k \leq P_{wt,\max}^k \\ 0 \leq P_{pv,t}^k \leq P_{pv,\max}^k \\ P_{wt,t}^k = \sum_{s_w=1}^S \pi_{t,s_w}^k P_{wt,t,s_w}^k \\ P_{pv,t}^k = \sum_{s_p=1}^S \pi_{t,s_p}^k P_{pv,t,s_p}^k \end{cases} \quad (21)$$

(5) Balance constraints of the DEL and DHL.

Based on the DR mechanism, the balance constraints for the DEL and DHL considering the load shift and interruption are as follows.

$$\begin{cases} P_{el,t}^k = \sum_{s_{el}=1}^S \pi_{t,s_{el}}^k P_{el,t,s_{el}}^k \\ Q_{hl,t}^k = \sum_{s_{hl}=1}^S \pi_{t,s_{hl}}^k Q_{hl,t,s_{hl}}^k \\ P_{ae,t}^k = P_{el,t}^k - P_{iel,t}^k - (P_{sne,t}^k - P_{spe,t}^k) \\ Q_{ah,t}^k = Q_{hl,t}^k - Q_{ihl,t}^k - (Q_{snh,t}^k - Q_{sph,t}^k) \end{cases} \quad (22)$$

(6) Dispatch constraints of the SEL and SHL.

In a planning horizon, loads shifted in and shifted out are balanced. Additionally, the shift in and out operations cannot be performed simultaneously. To mitigate the impact of the dispatch on electricity usage, the dispatch range and frequency are restricted by constraint (23)-(25).

$$\begin{cases} 0 \leq P_{spe,t}^k \leq U_{spe,t}^k P_{spe,\max}^k \\ 0 \leq P_{sne,t}^k \leq U_{sne,t}^k P_{sne,\max}^k \\ 0 \leq Q_{sph,t}^k \leq U_{sph,t}^k Q_{sph,\max}^k \\ 0 \leq Q_{snh,t}^k \leq U_{snh,t}^k Q_{snh,\max}^k \end{cases} \quad (23)$$

$$\begin{cases} \sum_{t=1}^T P_{spe,t}^k - \sum_{t=1}^T P_{sne,t}^k = 0 \\ \sum_{t=1}^T Q_{sph,t}^k - \sum_{t=1}^T Q_{snh,t}^k = 0 \end{cases} \quad (24)$$

$$\begin{cases} U_{spe,t}^k + U_{sne,t}^k \leq 1 \\ U_{sph,t}^k + U_{snh,t}^k \leq 1 \\ \sum_{t=1}^T (U_{spe,t}^k + U_{sne,t}^k) \leq N_{se} \\ \sum_{t=1}^T (U_{sph,t}^k + U_{snh,t}^k) \leq N_{sh} \end{cases} \quad (25)$$

where $U_{spe,t}^k$, $U_{sne,t}^k$, $U_{sph,t}^k$, and $U_{snh,t}^k$ are binary variables determining the transfer-in and transfer-out operations of SEL and SHL in VPPk at time t , respectively, 1 indicates the transfer of SEL and SHL, and 0 means no loads transferring.

(7) Dispatch constraints of the interruptible electric and heat load.

To ensure the rational response of IEL and IHL, the amount and frequency of load curtailments are limited as follows.

$$\begin{cases} 0 \leq P_{iel,t}^k \leq U_{iel,t}^k P_{iel,\max}^k \\ 0 \leq Q_{ihl,t}^k \leq U_{ihl,t}^k Q_{ihl,\max}^k \\ \sum_{t=1}^T U_{iel,t}^k \leq N_{iel} \\ \sum_{t=1}^T U_{ihl,t}^k \leq N_{ihl} \end{cases} \quad (26)$$

where $U_{iel,t}^k$ and $U_{sne,t}^k$ are binary variables determining the curtailment operations of IEL and IHL in VPPk at time t , respectively, 1 denotes the load interruption, and 0 means no loads being interrupted.

(8) Electrical and thermal balance constraints of the VPP.

Constraints (27) and (28) reveal the electrical and thermal equilibrium relationships for the VPP, which considers the internal generation, heating, energy storage, and energy transaction with the operator. Moreover, for the orderly energy transaction, the interaction between each VPP and the operator at the same period is unidirectional, and the volume of which is restricted.

$$\begin{cases} P_{vppe,t}^k = P_{vppeb,t}^k - P_{vppe,t}^k \\ Q_{vpph,t}^k = Q_{vpphb,t}^k - Q_{vpph,t}^k \end{cases} \quad (27)$$

$$\begin{cases} P_{ae,t}^k = P_{vppe,t}^k + P_{pv,t}^k + P_{wt,t}^k + P_{edis,t}^k - P_{ech,t}^k + P_{gt,t}^k \\ Q_{ah,t}^k = Q_{vpph,t}^k + Q_{gb,t}^k + Q_{whb,t}^k \\ 0 \leq P_{vppe,t}^k \leq U_{veb,t}^k P_{vo,max}^k \\ 0 \leq P_{vppe,t}^k \leq (1 - U_{veb,t}^k) P_{vo,max}^k \\ 0 \leq Q_{vpph,t}^k \leq U_{vhh,t}^k Q_{vo,max}^k \\ 0 \leq Q_{vpph,t}^k \leq (1 - U_{vhh,t}^k) Q_{vo,max}^k \end{cases} \quad (28)$$

where $U_{veb,t}^k$ and $U_{vhh,t}^k$ are binary variables that regulate the VPPk to make decisions on the power and heat trading with the operator at time t , respectively, 1 means the energy trading is carried out, and 0 means no energy trading.

3.3. Nash equilibrium proof of the Stackelberg game model

As mentioned above, the Stackelberg game is a decision-making process in which the operator and MVPP pursue their optimal objectives respectively. Taking the operator as the leader and MVPP as followers, the Stackelberg game model with one master and multiple slaves is constructed as follows, which contains the three elements of participants, strategies, and objective functions.

$$\begin{cases} G = \{\zeta; \lambda_{op}; \{\partial_{VPP1}, \partial_{VPP2}, \dots, \partial_{VPPk}\}; F_{op}; \{F_{VPP1}, F_{VPP2}, \dots, F_{VPPk}\}\} \\ \zeta = \{\text{operator}, VPP1, VPP2, \dots, VPPk\} \\ \lambda_{op} = \{\lambda_{vppe,t}, \lambda_{vppeb,t}, \lambda_{vpph,t}, \lambda_{vpphb,t}\} \\ \partial_{VPPk} = \{P_{vppeb,t}^k, P_{vppe,t}^k, Q_{vpphb,t}^k, Q_{vpph,t}^k, P_{edis,t}^k, P_{ech,t}^k, P_{gt,t}^k, Q_{gb,t}^k, Q_{whb,t}^k\} \end{cases} \quad (29)$$

where G represents the Stackelberg game model; ζ denotes the participants including the operator and MVPP; λ_{op} and ∂_{VPPk} denote the pricing strategy and the energy scheduling strategy for the operator and the VPPk, respectively.

After repeated iterations, the game reaches the Nash equilibrium when all followers make optimal decisions and the leader accepts these decisions. For the Nash equilibrium solution, participants on either side of the Stackelberg game cannot gain more benefits by unilaterally changing their strategies. Assume that $(\lambda_{op}^*, \partial_{VPP1}^*, \partial_{VPP2}^*, \dots, \partial_{VPPk}^*)$ is the Nash equilibrium solution of the Stackelberg game between the operator and MVPP in this paper, and the Nash equilibrium state of the game is as follows.

$$\begin{cases} F_{op}(\lambda_{op}^*, \partial_{VPP1}^*, \partial_{VPP2}^*, \dots, \partial_{VPPk}^*) \geq F_{op}(\lambda_{op}, \partial_{VPP1}^*, \partial_{VPP2}^*, \dots, \partial_{VPPk}^*) \\ F_{VPP1}(\lambda_{op}^*, \partial_{VPP1}^*, \partial_{VPP2}^*, \dots, \partial_{VPPk}^*) \leq F_{VPP1}(\lambda_{op}, \partial_{VPP1}^*, \partial_{VPP2}^*, \dots, \partial_{VPPk}^*) \\ F_{VPP2}(\lambda_{op}^*, \partial_{VPP1}^*, \partial_{VPP2}^*, \dots, \partial_{VPPk}^*) \leq F_{VPP2}(\lambda_{op}, \partial_{VPP1}^*, \partial_{VPP2}^*, \dots, \partial_{VPPk}^*) \\ \dots \\ F_{VPPk}(\lambda_{op}^*, \partial_{VPP1}^*, \partial_{VPP2}^*, \dots, \partial_{VPPk}^*) \leq F_{VPPk}(\lambda_{op}, \partial_{VPP1}^*, \partial_{VPP2}^*, \dots, \partial_{VPPk}^*) \end{cases} \quad (30)$$

For the theorem proving the existence and uniqueness of the Nash equilibrium solution achieved in the Stackelberg game, it can be divided into the following three aspects. a) The set of strategies of participants and followers is a non-empty tight convex set. b) There exists a unique optimal solution for all followers when the strategy of the leader is given. c) There exists a unique optimal solution for the leader when the strategies of followers are given.

According to the above three conditions, the existence and uniqueness of the Stackelberg equilibrium for the game model developed in this paper are proved as follows. a) From the model of the game participants, it can be seen that the strategy of the operator is required to satisfy the constraint (10), and the strategies of MVPP are restricted by constraint (18)–(28). Therefore, the strategy set of each participant is non-empty and tightly convex. b) The objective function of the VPP consists of four components. Among them, the cost functions for energy purchase and sale, DR compensation, and carbon trading are quasi-convex due to their linear characteristics. The cost function for the internal unit operation is non-linear, however, it is also quasi-convex as the quadratic term coefficients are positive. Thus, for given energy trading prices from the operator, the objective functions of MVPP are quasi-convex, and the unique optimal solution within the scope of the VPP strategy set exists. c) From formula (3)–(7), it is clear that the objective function of the operator is linear and quasi-convex. Therefore, for the energy scheduling strategy finalized by each VPP, the optimal energy trading prices determined by the operator within its strategy set are unique. In a nutshell, there exists a unique Nash equilibrium solution in the Stackelberg game model developed in this paper.

4. Stackelberg game solving based on the Kriging metamodel algorithm

As the lower-layer VPP optimal scheduling model established in this paper contains 0–1 variables, the optimization of the lower-layer model is not a convex optimization problem, and the optimal solution cannot be obtained by KKT conditions. Whereas the use of intelligent algorithms requires frequent calls to the model, which will affect the computational efficiency. It follows from the equilibrium solution of the Stackelberg game that each VPP develops the energy trading scheme based on the energy price signal. Therefore, there exists a mapping relationship between the trading price and the trading volume, and the mapping relationship in this paper is fitted by developing a computationally simplified Kriging metamodel, so as to replace the complex lower-layer model.

4.1. Development of the initial Kriging metamodel

Given that the CHP and GB fuel cost functions contain quadratic terms, the lower-layer model in this paper is a mixed integer nonlinear. Hence, the lower-layer model is required to be transformed into a mixed integer linear model through segment linearization, and solved by invoking the CPLEX solver on the MATLAB platform. The segment linearization procedure is specified in Appendix A. By substituting each set of energy prices into the VPP energy scheduling model developed above, the traded electricity and heat corresponding to transaction prices can be solved. Therefore, based on the mapping relationship between the trading price and trading volume, the lower-layer model is

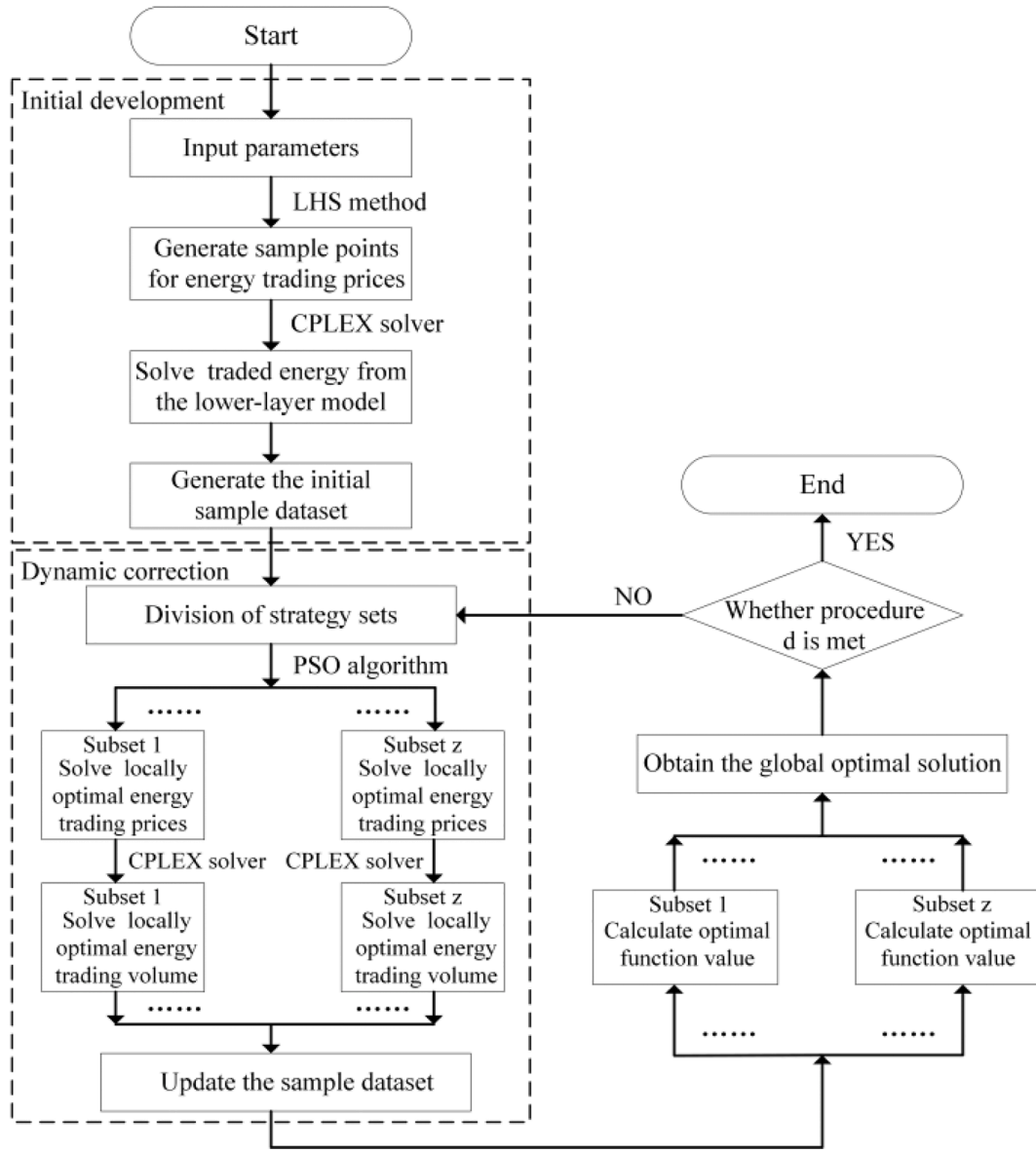


Fig. 5. Flowchart for solution.

converted into the initial Kriging metamodel and the two-layer game model can be transformed into a single-layer model, which are shown as formula (31) and (32) respectively.

$$\begin{cases} P_{vppe,t}^k = H(\lambda_{vppe,t}, \lambda_{vppeb,t}) \\ Q_{vpph,t}^k = H(\lambda_{vpph,t}, \lambda_{vpphb,t}) \\ (\lambda_{vppe,t}, \lambda_{vppeb,t}, \lambda_{vpph,t}, \lambda_{vpphb,t}) \in \Omega_{op} \end{cases} \quad (31)$$

$$\max F_{op}(\lambda_{vppe,t}, \lambda_{vppeb,t}, \lambda_{vpph,t}, \lambda_{vpphb,t}, P_{vppe,t}^1, Q_{vpph,t}^1, P_{vppe,t}^2, Q_{vpph,t}^2, \dots, P_{vppe,t}^k, Q_{vpph,t}^k) \quad (32)$$

where $H(\cdot)$ reflects the mapping between the trading price and trading volume; Ω_{op} is the strategy set space of the operator.

Considering that the Kriging metamodel requires a certain number of sampling points to be fitted, energy prices, as the input to the metamodel, are randomly sampled in this paper. Based on the LHS method, N_l sample points are generated for the purchase and sale prices of elec-

tricity and heat, denoted as $\lambda_{vppeb,j}$, $\lambda_{vppe,j}$, $\lambda_{vpphb,j}$, and $\lambda_{vpph,j}$, respectively. The elements contained are as follows.

$$\begin{cases} \lambda_{vppeb,j} = (\lambda_{vppeb,1,j}, \lambda_{vppeb,2,j}, \dots, \lambda_{vppeb,t,j}) \\ \lambda_{vppe,j} = (\lambda_{vppe,1,j}, \lambda_{vppe,2,j}, \dots, \lambda_{vppe,t,j}) \\ \lambda_{vpphb,j} = (\lambda_{vpphb,1,j}, \lambda_{vpphb,2,j}, \dots, \lambda_{vpphb,t,j}) \\ \lambda_{vpph,j} = (\lambda_{vpph,1,j}, \lambda_{vpph,2,j}, \dots, \lambda_{vpph,t,j}) \end{cases} \quad (33)$$

where j is the number of sample points generated; $\lambda_{vppeb,t,j}$, $\lambda_{vppe,t,j}$, $\lambda_{vpphb,t,j}$, and $\lambda_{vpph,t,j}$ denote the purchase and sale prices of electricity and heat at time t for the j th sample point, respectively.

The sample dataset for the metamodel which consists of transaction prices and the corresponding traded energy is represented as follows.

$$\begin{cases} D_j = (\lambda_{vppeb,j}, \lambda_{vppe,j}, \lambda_{vpphb,j}, \lambda_{vpph,j}, P_{vppe,j}^1, Q_{vpph,j}^1, P_{vppe,j}^2, Q_{vpph,j}^2, \dots, P_{vppe,j}^k, Q_{vpph,j}^k) \\ P_{vppe,j}^k = (P_{vppe,1,j}^k, P_{vppe,2,j}^k, \dots, P_{vppe,t,j}^k) \\ Q_{vpph,j}^k = (Q_{vppe,1,j}^k, Q_{vppe,2,j}^k, \dots, Q_{vppe,t,j}^k) \end{cases} \quad (34)$$

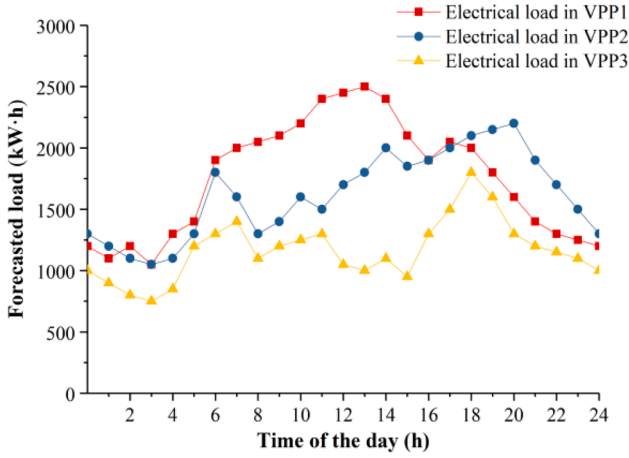


Fig. 6. Forecasted electric load in VPPs.

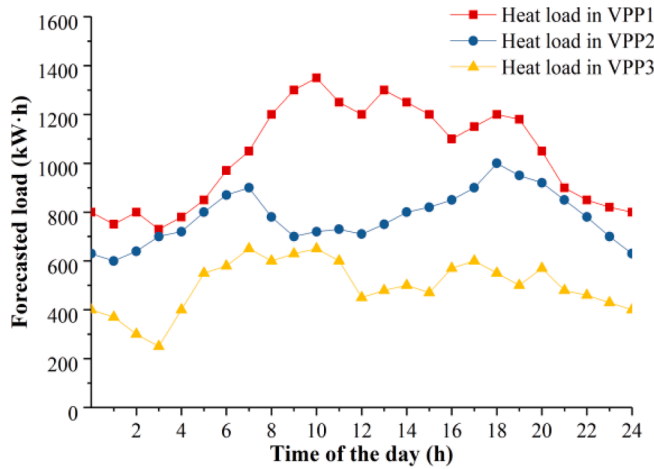


Fig. 7. Forecasted heat load in VPPs.

where D_j denotes the sample dataset; $P_{vpp,t,j}^k$ and $Q_{vpph,t,j}^k$ denote the traded electricity and heat at time t for the j th sample point.

4.2. Dynamic correction of the Kriging metamodel

The accuracy of the Stackelberg game solution depends on the Kriging model fit, which is directly influenced by the number of sample points, and excessive sample points can reduce the speed of the solution. In order to determine the proper number of sample points, this paper proposes a dynamic update algorithm of sampling points to modify the Kriging metamodel. Based on the initial Kriging model, the function values of the upper-layer pricing model can be derived. The strategy set in the upper layer is divided into multiple subsets according to the function values, thus eliminating the space of the strategy set containing a low probability of global optimal solutions and reducing the dynamic update range of sampling points. The strategy set division method is specified in Appendix B. For each strategy subset, the upper-layer model is solved with the particle swarm optimization (PSO) algorithm to obtain the locally optimal energy trading prices determined by the operator, and the lower-layer scheduling model is invoked to solve the corresponding locally optimal energy trading volume. Then, the solved locally optimal energy trading prices and volumes are added to the initial sample dataset, and the initial Kriging metamodel is modified to obtain the equilibrium solution of the Stackelberg game through iterations.

Based on the initial development and dynamic correction of the

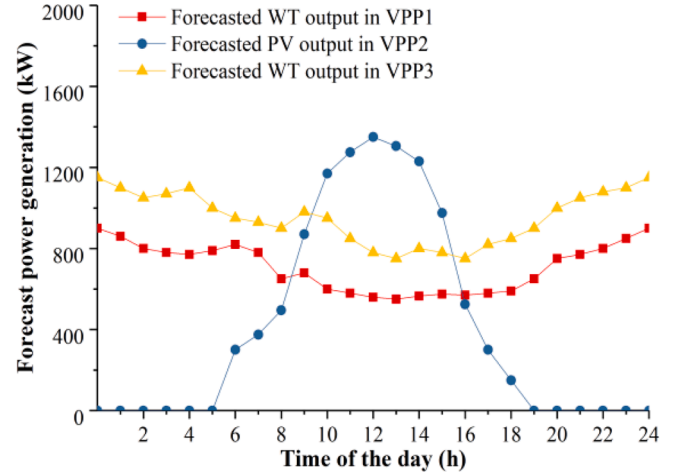


Fig. 8. Forecast power generation.

Kriging metamodel, the Stackelberg game model-solving process with the Kriging metamodel is shown in Fig. 5, and the detailed procedures are as follows.

- Input parameters for the operator and MVPP.
- Sample points for energy trading prices are generated based on the LHS, and the volume of energy traded for each VPP is solved by invoking the lower-layer energy scheduling model.
- Generate the initial sample dataset, and develop the initial Kriging metamodel to replace the lower layer model.
- Calculate the value of the objective function corresponding to each sample dataset, divide the strategy set into z subsets accordingly, and solve the locally optimal energy trading prices of each subset space with the PSO algorithm.
- Invoke the lower-layer model to solve the corresponding locally optimal energy trading volume and add them to the initial sample dataset.
- Based on the updated sample dataset, the initial Kriging metamodel is corrected, and the optimal function value of the upper-layer model in each subset is calculated.
- Compare the optimal function values in all subsets and take the maximum value as the global optimal solution for the current iteration.
- Determine whether the global optimal solutions of the two iterations are equal, if not, continue iterating, otherwise output the final result.

5. Case study

5.1. Basic data

In this paper, a typical energy-using region in northern China is selected and three representative VPPs, namely VPP1, VPP2, and VPP3, are chosen to verify the effectiveness of the proposed Stackelberg game model and carbon trading mechanism. The platform for the development of the Kriging metamodel is the DACE toolbox in MATLAB. VPP1 and VPP2, which are configured for distributed wind power and rooftop PV respectively, are located in urban areas with high energy demand. Whereas the VPP3 is located in a suburban area and equipped with a high-capacity wind farm. In addition, it is assumed that the gas supply is sufficient for each VPP to meet the maximum output demands of the CHP and GB. To avoid large-scale centralized energy trading, the maximum amount of electricity and heat allowed to be traded to the operator per hour is set as 800 kW and 500 kW respectively. Figs. 6 and 7 show the forecasted electric and heat load curves in each VPP. Fig. 8 shows the forecasted output curves for the WT and PV. The curves reflecting the range of the power grid tariff, feed-in tariff, and thermal

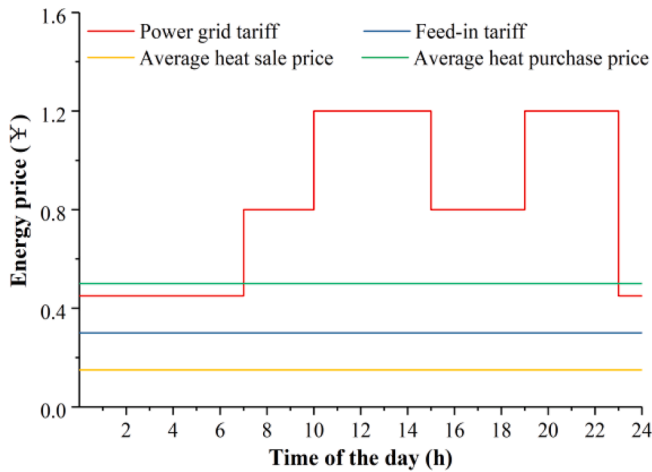


Fig. 9. Energy prices for the operator and MVPP.

Table 1

Composition of four cases ($\checkmark$ = components included, $\times$ = components excluded).

Case	Stackelberg game	traditional carbon trading	ladder-type carbon trading	traditional DR	dual-compensated DR
1	$\checkmark$	$\times$	$\checkmark$	$\times$	$\checkmark$
2	$\times$	$\times$	$\checkmark$	$\times$	$\checkmark$
3	$\checkmark$	$\checkmark$	$\times$	$\times$	$\checkmark$
4	$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\times$

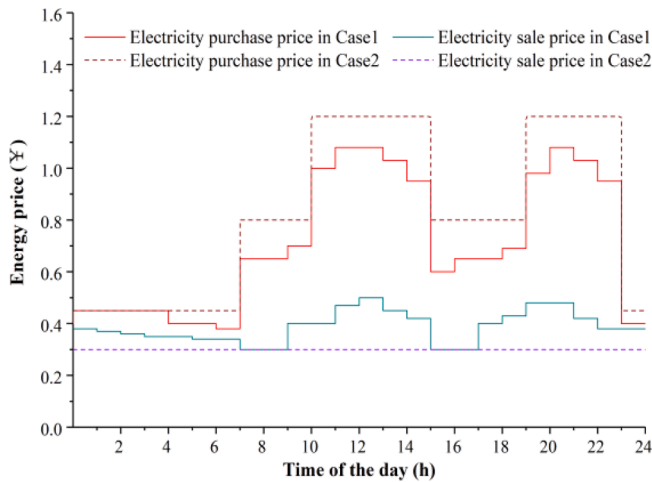


Fig. 10. Electricity trading prices for VPPs.

energy trading are shown in Fig. 9. Table D1 in Appendix D presents the specific parameters for each VPP internal unit including the CHP, GB, and ESS. Tables D2 and D3 in Appendix D provide the relevant parameters for carbon trading and DR respectively.

5.2. Results analysis

Four cases designed in this paper to analyze various aspects of the Stackelberg game model for energy trading, the pricing mechanism for carbon trading, and the compensation form for DR are shown in Table 1. Case 1 considers the mechanisms for the Stackelberg game, the ladder-type carbon trading, and the dual-compensated DR. With the same carbon trading and DR mechanism, the Stackelberg game model is not

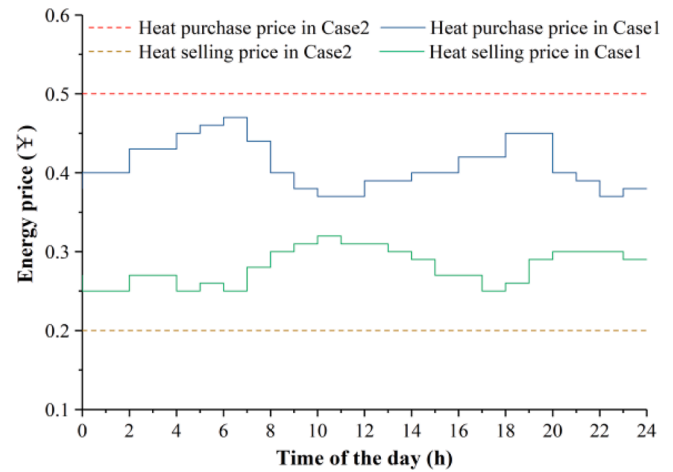


Fig. 11. Heat trading prices for VPPs.

considered in Case 2, and the operator adopts market trading prices to carry out energy transactions with each VPP. With the same Stackelberg game structure as Case 1, Case 3 adopts the traditional carbon trading mode with a fixed carbon price, and Case 4 uses the traditional single-compensated DR mechanism.

5.2.1. Analysis of electricity-heat trading results

Comparing Case 1 with Case 2, the pricing strategies for the electricity and heat trading are respectively shown in Figs. 10 and 11. It can be seen that the energy trading prices in Case 1 fluctuate within the upper and lower limits. As the Stackelberg game model proposed in this paper is not considered in Case 2, the power grid tariff and feed-in tariff, as well as the average heat purchase and selling price, are taken as the energy trading prices for each VPP in Case 2. Fig. 12 shows the power transactions between VPPs and the operator, and the positive and negative sides of the power trading represent the purchase and sale decisions for each VPP respectively. Fig. 13 shows the internal power exchange between VPPs, which is mediated by the operator.

Combining Figs. 10, 12, and 13, the game results of the power trading price and volume are analyzed in this paper. During hours 1–4, all VPPs purchase power from the operator, without the engagement of internal power exchange. To ensure revenue, the operator sets the electricity purchase price as the power grid tariff. During hours 8–9 and 16–17, all VPPs sell electricity to the operator, and similarly, the selling price is set as the feed-in tariff to guarantee costs. In periods 5–7, 14–15, and 22–24, the overall manifestation of the three VPPs is the sale of electricity to the operator as there is still a surplus after the internal power trading. Accordingly, considering the lower feed-in tariff, the power purchase price decreases to attract VPPs to increase their power purchases during certain periods, thus reducing power sales to the upper grid. Therefore, comparing Case 1 with Case 2, the obvious increases in the power purchases of VPP1 and VPP2 are respectively concentrated in hours 14–15 and 22–24. In the time periods 10–13 and 18–21, VPPs make different power trading decisions, with an overall performance of purchasing power from the operator as the internal power exchange cannot meet the electricity demand. During these periods, the electricity selling price for VPPs rises correspondingly, so as to induce VPPs to increase electricity sales by optimizing the internal unit outputs. As a result, there is a significant increase in electricity sales for VPP3 especially over the time periods 19–21 in Case 1 compared to that in Case 2.

As can be seen in Fig. 13, both the volume and frequency of the internal power exchange significantly increase in Case 1 compared to those in Case 2. It can be attributed to the increase in the power trading of VPPs during partial time periods under the guidance of the price signal to prioritize internal power interactions, thus reducing transactions with the upper grid. Fig. 14 shows the optimal power dispatch

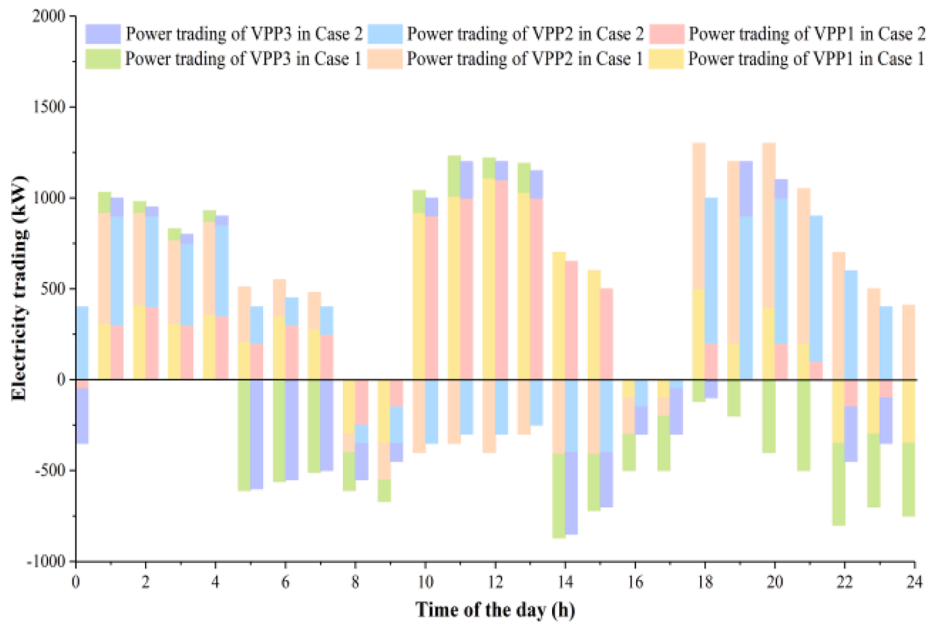


Fig. 12. Power trading between VPPs and the operator.

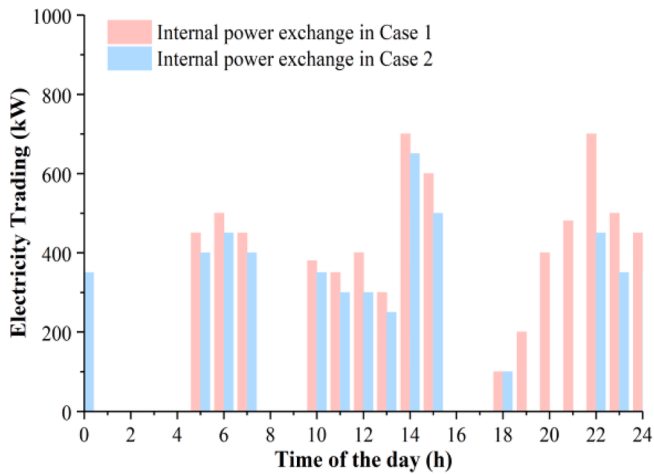


Fig. 13. Internal power exchange.

for each VPP in Case 1 and 2. For VPP1, the GT output and the load curtailment are both visibly reduced in Case 1 in comparison with those in Case 2 during hours 18–19, which can be attributed to the increase in power purchase for the lower purchase price during the period. Due to the increase in the power sale caused by the higher selling price during hours 22–24, a significant increase can be seen in the GT output. Similar to VPP1, VPP2 and 3 adjust the scheduling scheme for DERs and loads in response to the optimal power trading price issued by the operator.

Fig. 15 shows the heat transactions between VPPs and the operator, and Fig. 16 shows the internal heat exchange between VPPs coordinated by the operator. Combining Figs. 15 and 16, and Fig. 11, it can be seen that the internal heat exchange in Case 1 is concentrated in the periods 9–12 and 20–21 when the heat purchase price is lower and the heat selling price is higher. Fig. 17 shows the optimal heat dispatch for each VPP in Case 1 and 2. As can be seen in Fig. 17, internal units within each VPP can meet its heating demand in most periods, and the heat shortage during the peak period can be supplemented by internal thermal energy trading. Comparing Case 1 with Case 2, the curtailment and shift amount of heat load in VPP1 are both reduced. The reason for the reduction is that the heat trading price is lower than the DR compensation prices of

IHL and SHL. Therefore, VPP1 prioritizes the purchase of heat from VPPs with surplus heat resources, which accounts for the increase in internal heat transactions in Case 1.

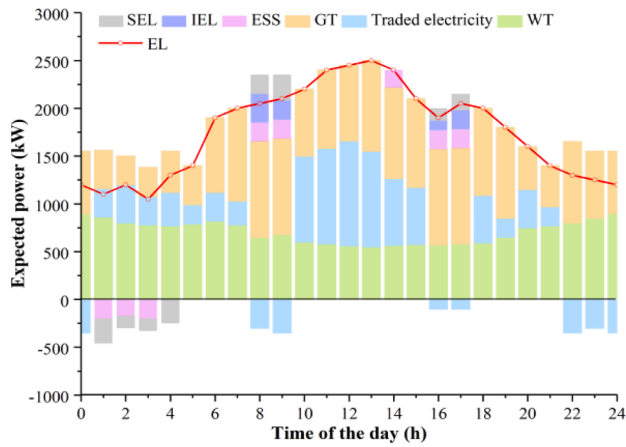
5.2.2. Analysis of the Stackelberg game process

Fig. 18 shows the Kriging metamodel solving process of the Stackelberg game between the operator and MVPP. As shown in Fig. 18, the Kriging metamodel-based method has good convergence. As the number of iterations increases, there are different convergence trends for the game participants, manifesting as an increase in the profit of the operator and decreases in the operating costs of MVPP, which eventually reach a joint convergence at the 45th iteration. After 45 iterations, the game equilibrium is reached between the operator and MVPP, with each participant unable to reap more benefits by adjusting their strategy individually.

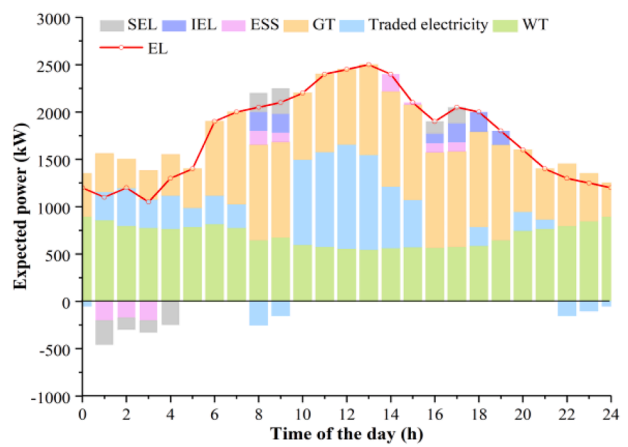
Table 2 is a comparative analysis of Case 1 and 2, which shows the changes in costs of MVPP and the profit of the operator before and after the Stackelberg game. As shown in Table 2, the costs for VPPs in case 1 are higher than that in case 2, mainly because the optimal energy trading prices, which are more economical than market prices, increase the volume of internal energy exchange. Moreover, the cost of VPP3 is reduced by the most after the Stackelberg game, at 13.89 %, exceeding 5.22 % of VPP1 and 6.40 % of VPP2. Considering the lower energy demand within VPP3, the significant decline in VPP3 cost can be attributed to a greater proportion of energy resources being involved in the internal transaction. Compared to Case 2, the profit for the operator rises by 11.07 % in Case 1, the main reason is that the increased internal energy exchange decreases the transaction volume from the upper grid, which reduces the power purchase cost and increases the power sale revenue.

5.2.3. Analysis of carbon trading results

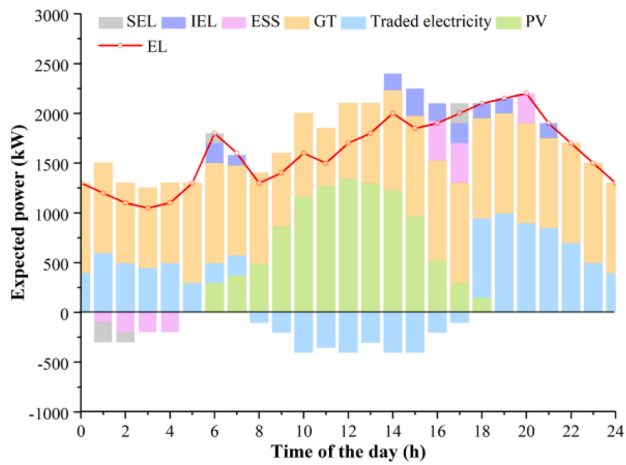
Taking VPP1 in Case 2 as an example, this paper performs the sensitivity analysis of the benchmark carbon price, the carbon price growth rate, and the interval length, and discusses the impact of changes in these parameters on carbon emissions and carbon trading costs. Fig. 19 (a) presents the sensitivity of the benchmark carbon price. For benchmark carbon price less than 55 ¥/t, the constraint on the cost of carbon trading increases with the price, so VPP1 reduces carbon emissions by adjusting unit output and load distribution. However, when the benchmark carbon price is greater than 55 ¥/t, the outputs of energy



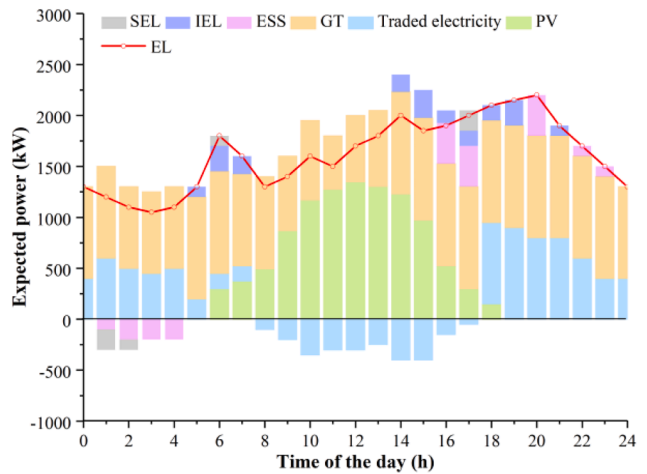
(a) Power dispatch for VPP1 in Case 1



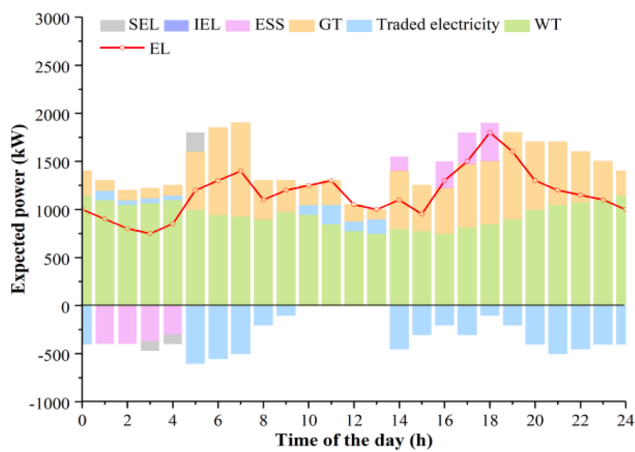
(b) Power dispatch for VPP1 in Case 2



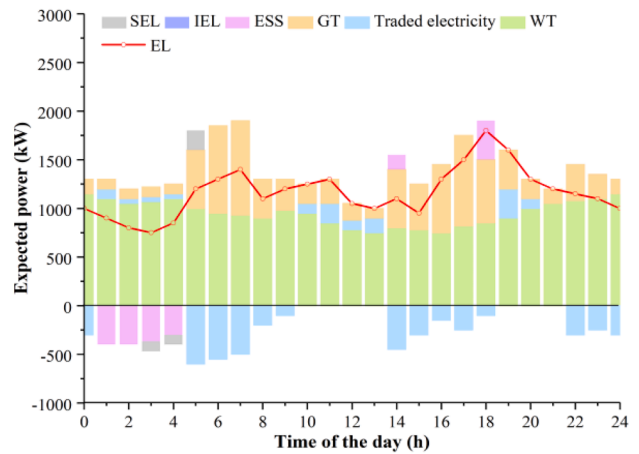
(c) Power dispatch for VPP2 in Case 1



(d) Power dispatch for VPP2 in Case 2



(e) Power dispatch for VPP3 in Case 1



(f) Power dispatch for VPP3 in Case 2

Fig. 14. Power dispatch for VPPs.

resources within the VPP1 tend to be stable, which results in carbon emissions being little affected by changes in the price. The sensitivity of the carbon price growth rate is shown in Fig. 19 (b). It can be seen that with increasing price growth, the cost of carbon trading increases while carbon emissions decrease. Eventually, carbon emissions level off when the price growth exceeds 0.2. It can still be attributed to the limited

regulating capacity of the internal units, thus constraining the ability of carbon trading to restrict carbon emissions. Fig. 19(c) shows the sensitivity for the length of the interval. When the interval length is less than 2 t, VPP1 purchases CEAs at higher gradient prices, resulting in the higher carbon trading cost and lower carbon emission. There exist significant changes in the carbon emission and carbon trading cost within

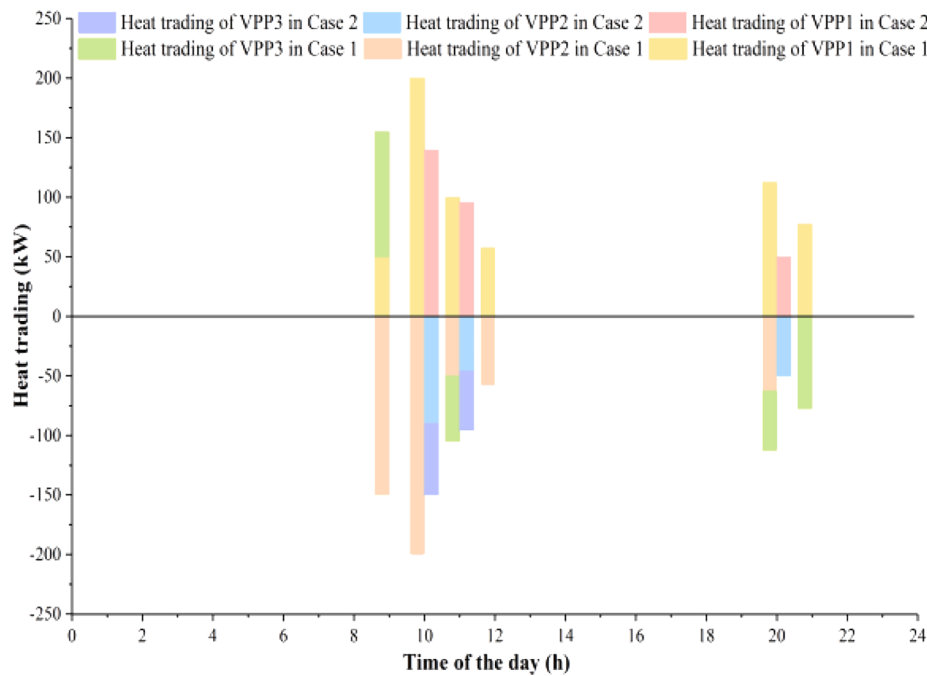


Fig. 15. Heat trading between VPPs and the operator.

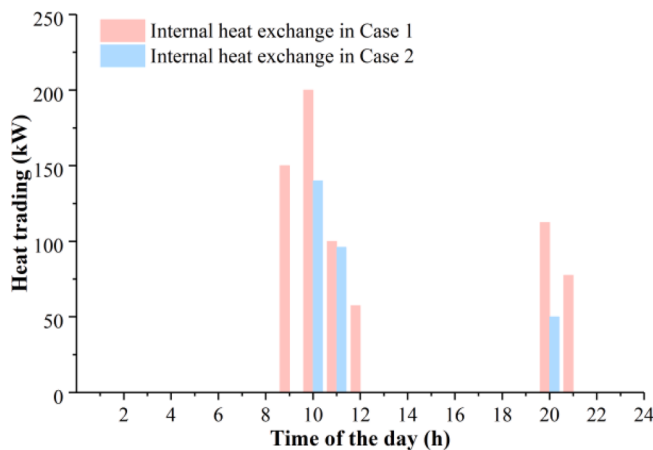


Fig. 16. Internal heat exchange between VPPs.

the range of 2 to 3 t interval length. As the interval length increases, fewer CEAs are purchased at prices of higher gradients, which accounts for the reduced carbon trading cost and increased carbon emission. When the interval length exceeds 3 t, CAEs are mostly traded at the price of the first gradient, so both the carbon trading cost and emission reach stability. Therefore, in order to balance the carbon emission and carbon trading cost of the VPP, the reasonable parameter setting of the ladder-type pricing mechanism is highly demanding.

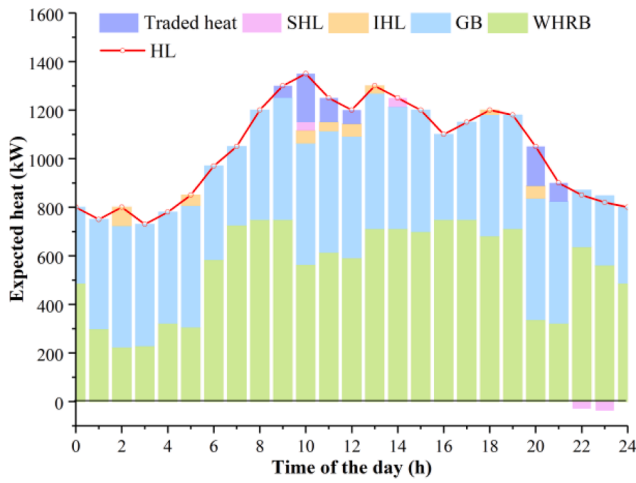
Considering the limitations of parameter regulation in the ladder-type carbon trading mechanism, the changes in carbon emissions before and after the Stackelberg game is further examined in this paper. Table 3 is a comparative analysis of Case 1 and 2 in terms of low-carbon performance. For the carbon emission volume and the carbon trading cost after the Stackelberg game, decreases can be seen in VPP1 and 2, while an increase can be seen in VPP3. This is because, compared to Case 2, VPP1 and 2 in Case 1 are actively involved in internal energy trading under the price guidance, and purchase energy from other entities during peak energy consumption hours, enabling the optimization of distributed energy and load dispatch. VPP3, due to its more abundant

internal energy resources, can obtain incentives from the carbon trading market before the Stackelberg game. After the Stackelberg game, VPP3 sells more energy to other entities mainly by increasing the unit outputs for some periods, which leads to an increase in carbon emissions. Therefore, internal energy trading is essentially accompanied by the trading of CEAs between VPPs. The VPP profiting from the carbon trading market can transfer the excess CEA to other entities in the form of internal energy trading, promoting the overall carbon emission balance.

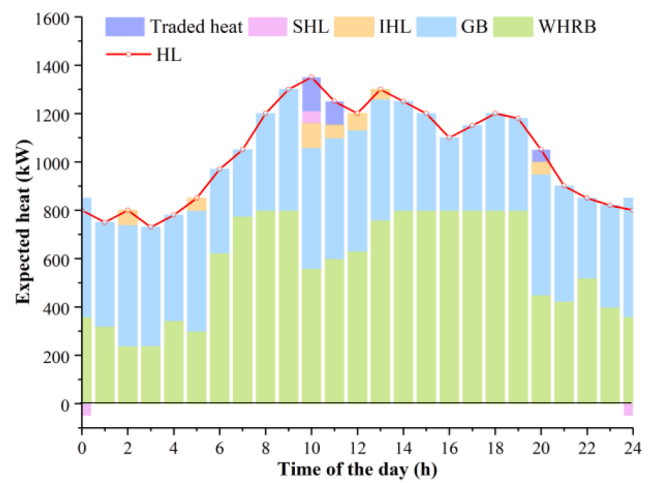
Table 4 compares carbon emissions and carbon trading costs under a traditional fixed carbon price mechanism. Comparing Case 1 with Case 3, the reduction can be seen in carbon emissions from VPP1, 2, and 3, by 4.20 %, 4.17 %, and 3.51 % respectively, whereas the carbon trading costs for VPP1 and 2 increase slightly. The main reason is that the ladder-type carbon price mechanism achieves a graded optimization of carbon trading by refining carbon trading bands and prices, which effectively restrains carbon emissions compared to the rough regulation of the traditional mechanism. As the ladder-type carbon price in the corresponding band tends to be higher than the traditional fixed carbon price, it may slightly increase the cost of carbon trading.

5.2.4. Analysis of DR results

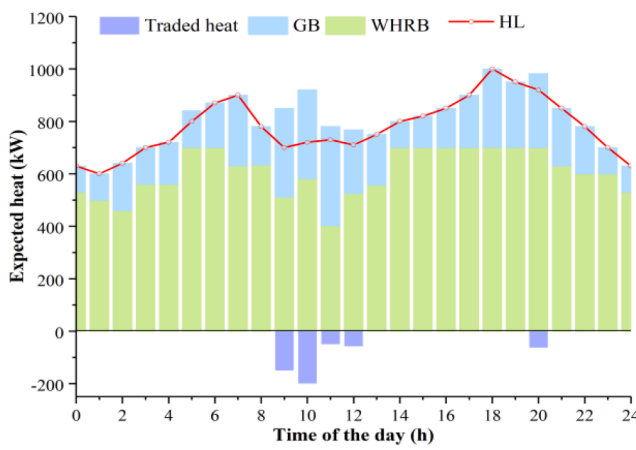
In this paper, the dual-compensated mechanism for DR consists of both the energy price compensation and the carbon subsidy, while the traditional DR in Case 4 only considers the compensation for energy prices. Table 5 is a comparative analysis of Case 1 and 4, which show the associated costs and carbon emissions under the two mechanisms. Comparing Case 1 with Case 4, increased price compensation costs and decreased carbon emission can be seen for the three VPPs. The main reason for the increase is that the dual compensation mechanism motivates VPPs to reduce energy usage during peak hours, and the outputs of the units during the corresponding hours are reduced. Compared to other VPPs, VPP1 sees the most changes in price compensation costs and carbon emissions, with variations of 6.39 % and 3.25 % respectively. It can be attributed to the greater load curtailments of VPP1, especially for thermal loads. Therefore, loads within VPP1 are more adjustable and more sensitive to the dual compensation mechanism. In addition, as the reduction in carbon trading costs caused by the dual compensation mechanism can offset the incremental costs of the DR compensation,



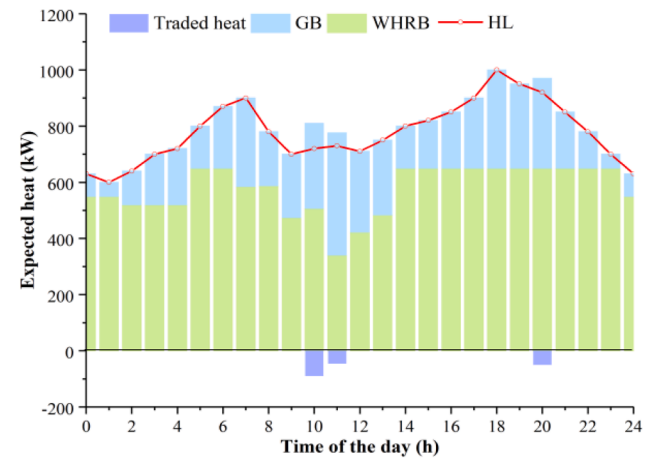
(a) Heat dispatch for VPP1 in Case 1



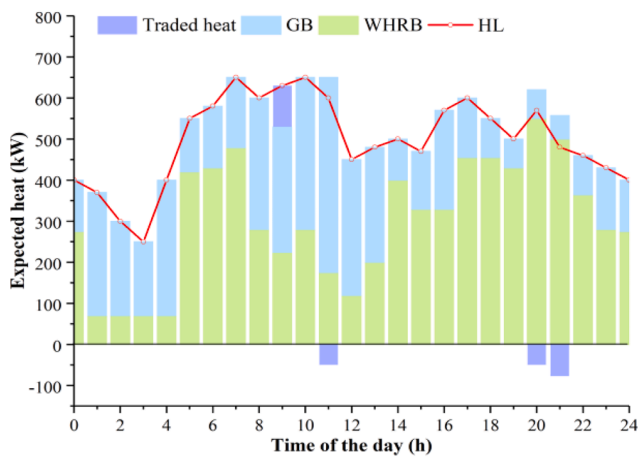
(b) Heat dispatch for VPP1 in Case 2



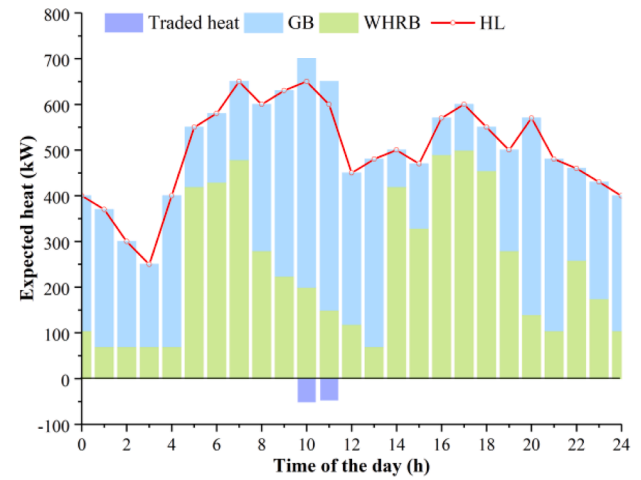
(c) Heat dispatch for VPP2 in Case 1



(d) Heat dispatch for VPP2 in Case 2



(e) Heat dispatch for VPP3 in Case 1



(f) Heat dispatch for VPP3 in Case 2

Fig. 17. Heat dispatch for VPPs.

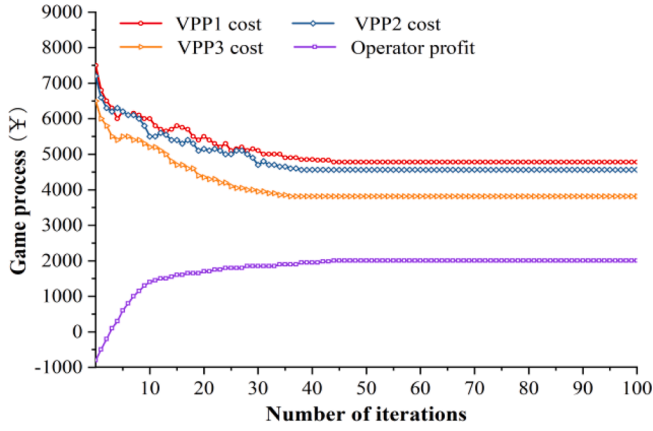


Fig. 18. Convergence process of Stackelberg game.

Table 2

Economic analysis of the Stackelberg game.

Case	VPP1 cost (¥)	VPP2 cost (¥)	VPP3 cost (¥)	operator profit (¥)
1	4778	4562	3814	2005
2	5041	4874	4429	1783

there will be little impact on the total cost of the VPP.

6. Conclusion

Based on a ladder-type carbon price mechanism for carbon trading and a dual compensation mechanism for DR, this paper develops a multi-stakeholder Stackelberg game strategy with the operator as the leader and VPPs as followers. Considering the multi-energy coupling characteristic, the strategy realizes the optimal low-carbon and economic scheduling for the operator and VPPs. Multiple cases are set up for comparative analysis, and the conclusions are as follows.

- a) In the Stackelberg game model, the pricing strategy from the operator guides MVPP to dispatch internal resources and carry out energy trading rationally, while the energy purchase and sale decisions for MVPP will in turn influence the setting of energy prices. With the optimization of energy prices, the Stackelberg game strategy facilitates the internal energy exchange between VPPs, which further promotes energy consumption within the VPP and coordinates the complementarity of energy resources from other VPPs. As a result, the total cost of a VPP can be reduced by up to 13.89 %, while the profit of the operator is increased by 11.07 %.

Table 3

Low-carbon performance analysis of Stackelberg game.

Cases	VPP number	Carbon emission (t)	Carbon trading cost (¥)
1	VPP1	9.83	468
	VPP2	7.81	376
	VPP3	6.23	-215
2	VPP1	10.47	514
	VPP2	8.32	412
	VPP3	5.65	-254

Table 4

Comparison of carbon trading mechanisms.

Cases	VPP number	Carbon emission (t)	Carbon trading cost (¥)
3	VPP1	10.25	461
	VPP2	8.15	367
	VPP3	6.56	-201

Table 5

Comparison of DR mechanisms.

Cases	VPP number	DR compensation cost (¥)	Carbon emission (t)	Carbon trading cost (¥)
1	VPP1	219	9.83	468
	VPP2	172	7.81	376
	VPP3	102	6.23	-215
4	VPP1	205	10.16	481
	VPP2	161	8.01	385
	VPP3	96	6.35	-207

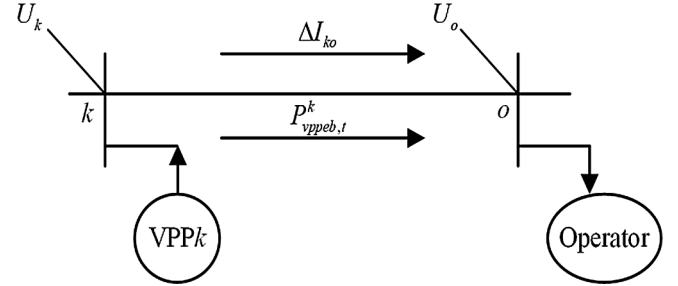
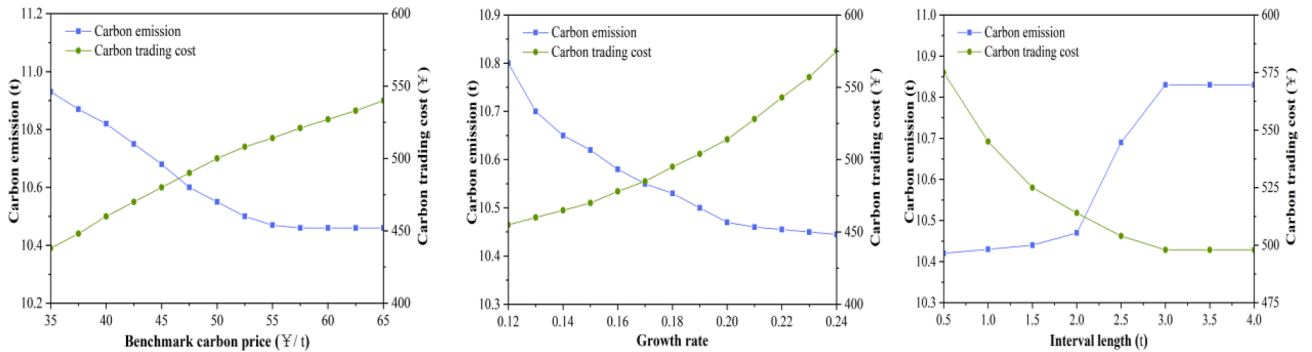


Fig. A1. Power transmission of VPP k.

Table A1

Relevant parameters of the energy transmission.

Parameter	$r_{10}(\Omega)$	$r_{20}(\Omega)$	$r_{30}(\Omega)$	$\kappa(\text{¥/kWh})$	$\alpha_{th}(\text{¥/kWh})$
Value	5.12	4.81	4.73	0.02	0.03



(a) Sensitivity of benchmark price (b) Sensitivity of growth rate (c) Sensitivity of interval length

Fig. 19. Sensitivity of carbon trading parameters.

Table D1
Operating parameters of internal units.

	Parameter	Unit	Value
GTs	Ramping rate	kW/h	500
	Descending rate	kW/h	500
	Maximum output	kW	1000
	Efficiency	—	0.35
	α_e^k	—	0.15
	b_e^k	—	0.5
	c_e^k	—	0.8
WHRB	Efficiency	—	0.83
GBs	Ramping rate	kW/h	200
	Descending rate	kW/h	200
	Maximum output	kW	500
	Efficiency	—	0.45
	α_h^k	—	0.25
	b_h^k	—	0.35
	c_h^k	—	1.2
ESS	Maximum capacity	kWh	1000
	Minimum capacity	kWh	200
	Maximum discharge quantity	kW	500
	Maximum charge quantity	kW	500
	Discharging efficiency	—	0.9
	Charging efficiency	—	0.9
	Maximum operation number	—	12
	Rental cost	¥/kWh	0.2

Table D2
Relevant parameters for the carbon trading.

Parameter	$v(\text{¥/t})$	λ_c	$\mu(\text{kg})$	$\chi_e(\text{kg/kWh})$	$\chi_h(\text{kg/kWh})$	$\varepsilon_c(\text{kg/kWh})$	σ_c
Value	55	0.2	2000	0.798	0.385	1.054	0.7

Table D3
Relevant parameters for DR.

VPP1		VPP2		VPP3	
Parameter	Value	Parameter	Value	Parameter	Value
$\alpha_{drie}[\text{¥/(kWh)}]$	0.2	$\alpha_{drie}[\text{¥/(kWh)}]$	0.2	$\alpha_{drie}[\text{¥/(kWh)}]$	0.3
$\alpha_{drih}[\text{¥/(kWh)}]$	0.45	$\alpha_{drih}[\text{¥/(kWh)}]$	0.45	$\alpha_{drih}[\text{¥/(kWh)}]$	0.5
$\alpha_{drse}[\text{¥/(kWh)}]$	0.2	$\alpha_{drse}[\text{¥/(kWh)}]$	0.2	$\alpha_{drse}[\text{¥/(kWh)}]$	0.25
$\alpha_{drsh}[\text{¥/(kWh)}]$	0.45	$\alpha_{drsh}[\text{¥/(kWh)}]$	0.45	$\alpha_{drsh}[\text{¥/(kWh)}]$	0.5
τ_{se}	0.2	τ_{se}	0.2	τ_{se}	0.2
τ_{sh}	0.3	τ_{sh}	0.3	τ_{sh}	0.3
N_{se}	10	N_{se}	10	N_{se}	12
N_{sh}	10	N_{sh}	10	N_{sh}	12

b) In comparison to regulating the benchmark value and growth rate of the ladder-type carbon price, the Stackelberg game strategy, which

promotes the CEA trading, can further optimize energy resource outputs, thereby fully exploiting the carbon reduction potential of MVPP and further constraining carbon emissions. Specifically, the carbon emission of a VPP can be reduced by up to 4.20 %.

c) Compared to the rough regulation of the traditional carbon trading model, the ladder-type carbon price mechanism achieves a graded optimization of carbon trading by refining carbon trading bands and prices, which provides a better motivation to reduce carbon emissions. Whereas the dual-compensated DR mechanism optimizes the load distribution by quantifying the contribution of load curtailments to the carbon reduction.

With the ongoing development of VPP, greater requirements have been put forward for the research on the economical and low-carbon operation. In this paper, the optimal electricity-heat-carbon dispatch for MVPP considering the Stackelberg game is only preliminarily studied. There is still a lot of research space, and some aspects are simplified for the limited ability and time, especially for the safety and reliability of the VPP operation. In future research, we will analyze the energy flow constraints based on the topology of the power and heat networks, and investigate the congestion management for energy transmission and distribution.

CRedit authorship contribution statement

Jinye Cao: Methodology, Writing – original draft. **Dechang Yang:** Conceptualization, Project administration, Writing – review & editing. **Payman Dehghanian:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgment

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Appendix A

See Fig. A1. And Table A1.

Fig. A1 shows the power transmission between VPPk and the operator on the distribution side. It is assumed that the nodal voltage levels of VPPk and the operator are relatively stable, which are close to the nominal voltage U_N , and VPPk sells electricity to the operator. The incremental current ΔI_{ko} generated in the line by the power purchased by the operator can be expressed as formula (A1).

$$\Delta I_{ko} \approx \frac{P_{vppes,t}^k}{U_N} \quad (\text{A1})$$

Ignoring the line impedance and considering only the line resistance r_{ko} , the power losses ΔP_{loss}^k arising from electrical energy trading can be approximated as formula (A2).

$$\Delta P_{loss}^k \approx \Delta I_{ko}^2 r_{ko} = \frac{r_{ko}}{U_N^2} (P_{vppes,t}^k)^2 \quad (\text{A2})$$

The power transmission cost shared by VPPk is calculated as formula (A3).

$$C_{te,t}^k = \frac{1}{2} \kappa \frac{r_{ko}}{U_N^2} (P_{vppes,t}^k)^2 \quad (A3)$$

where $C_{te,t}^k$ denotes the power transmission cost of VPPk, κ denotes the cost of electricity compensation for unit power loss.

In this paper, α_{te}^k is set as the conversion coefficient of the VPPk power transmission cost, and the specific expression of which is as follows.

$$\alpha_{te}^k = \kappa \frac{r_{ko}}{U_N^2} \quad (A4)$$

Similarly, it can be inferred that the transmission cost brought about by the power purchase of VPPk. As with the power transmission cost, the transmission cost of heat is related to transmission losses, whereas the transmission losses in the thermal network include water pressure losses and pipe temperature losses, which are complicated to model mathematically. To simplify the calculations, drawing on reference [37], this paper approximates the heat transmission cost and heat transmission amount as a linear relationship, which is specified as follows.

$$C_{th,t}^k = \frac{1}{2} \alpha_{th} |Q_{vpphb,t}^k - Q_{vpphs,t}^k| \quad (A5)$$

where $C_{th,t}^k$ denotes the heat transmission cost of VPPk, α_{th} is the conversion coefficient of the VPP heat transmission cost.

Table A1 presents the relevant parameters for energy transmission. Considering the power and heat transmission cost, the total transmission cost of VPPk is shown in formula (A6).

$$C_{tran,t}^k = \frac{1}{2} (\alpha_{te}^k |P_{vppes,t}^k - P_{vppes,t}^k|^2 + \alpha_{th} |Q_{vpphb,t}^k - Q_{vpphs,t}^k|) \quad (A6)$$

Appendix B

Formula (12) and (13) are the fuel cost functions for which a segmented linearization is applied as follows.

- (a) Based on the required accuracy, $Q+1$ segmentation points $e_1, e_2, \dots, e_{Q+1}$ are taken, dividing the original function into Q intervals.
- (b) $Q+1$ continuous auxiliary variables $w_1, w_2, \dots, w_{Q+1}$ and Q binary auxiliary variables $u_1, u_2, \dots, u_{Q+1}$ are set to satisfy the following equation.

$$\begin{cases} w_1 + w_2 + \dots + w_{Q+1} = 1 \\ u_1 + u_2 + \dots + u_Q = 1 \\ w_1 \geq 0, w_2 \geq 0, \dots, w_{Q+1} \geq 0 \\ w_1 \leq u_1, w_2 \leq u_1 + u_2, \dots, w_Q \leq u_{Q-1} + u_Q, w_{Q+1} \leq u_Q \end{cases} \quad (B1)$$

- (c) The non-linear function in formula (12) is replaced with the following linear expressions.

$$\begin{cases} P_{gt,t}^k = \sum_{q=1}^{Q+1} w_q e_q \\ C_{fc,t}^k = \sum_{q=1}^{Q+1} w_q C_{fc,t}^k(e_q) \end{cases} \quad (B2)$$

Similarly, formula (13) is linearized through formula (B1) and (B2).

Appendix C

The details of the strategy set division method are as follows.

- (a) The initial sample data set D_j is substituted into the objective function of the upper-layer pricing model and the corresponding function values are calculated as $F_{VPP0,j}$, which consist of the set V as (C1).

$$V = \{(\lambda_{vppeb,j}, \lambda_{vppej}, \lambda_{vppeb,j}, \lambda_{vppej}, F_{VPP0,j}) | j = 1, 2, \dots, N_l\} \quad (C1)$$

- (b) Given that z is the number of the upper-layer strategy subset, the maximum objective function value and the corresponding energy trading price are selected in the set V .

$$\begin{cases} f_z = \max F_{VPP0,j} \\ (\lambda_{vppeb,z}, \lambda_{vppe,z}, \lambda_{vppeb,z}, \lambda_{vppe,z}) = \arg\max F_{VPP0,j} \end{cases} \quad (C2)$$

where f_z is the maximum value of the upper-layer objective function in the z th strategy subset; $\lambda_{vppeb,z}$, $\lambda_{vppe,z}$, $\lambda_{vppeb,z}$, and $\lambda_{vppe,z}$ denote the locally optimal purchase and sale prices for the electricity and heat, respectively.

(c) Ranges of the strategy subsets are respectively circled with $\lambda_{vppeb,z}, \lambda_{vppe,z}, \lambda_{vppeb,z}$, and $\lambda_{vppe,z}$ as centers, and r_1, r_2, r_3 , and r_4 as radiuses. ($\lambda_{vppeb,z}, \lambda_{vppe,z}, \lambda_{vppeb,z}, \lambda_{vppe,z}, F_{VPO,j}$) within the range is included in the set V_z . Radiuses are calculated as (C2) and (C3).

$$r_{1,2} = \frac{|\lambda_{gb,max} - \lambda_{gs,min}|(I_{max} - I + 1)}{3I_{max}} \quad (C3)$$

$$r_{3,4} = \frac{|\lambda_{h,max} - \lambda_{h,min}|(I_{max} - I + 1)}{3I_{max}} \quad (C4)$$

where $\lambda_{gb,max}$ and $\lambda_{gs,min}$ denote the maximum and minimum electricity trading prices, respectively; I and I_{max} denote the current number and the maximum number of iterations, respectively.

(d) For the remainder of the set V , step (b) and (c) are continued until the end of the set division.

Appendix D

See Tables D1–D3.

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